

AMC Module

Block Copolymer Nanotechnology

Faculty of Mathematics and Natural Sciences

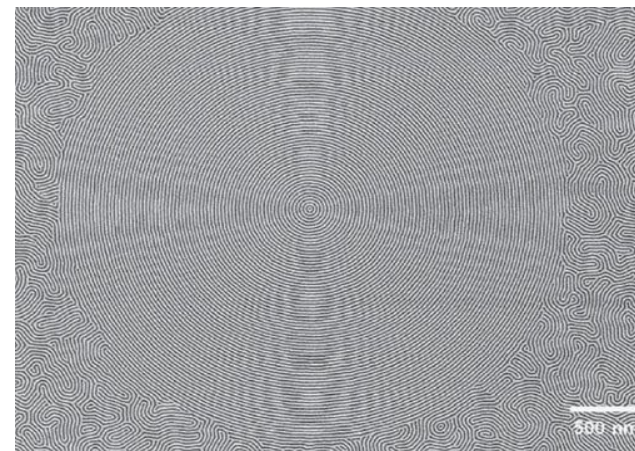
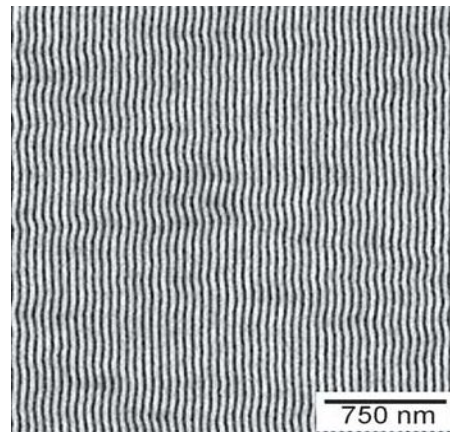
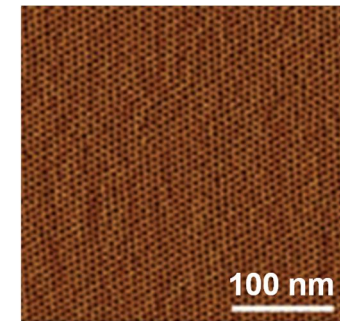
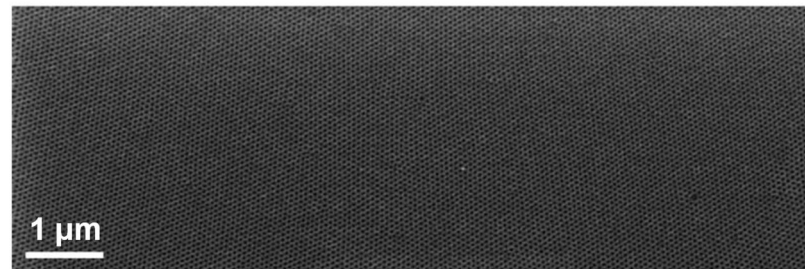
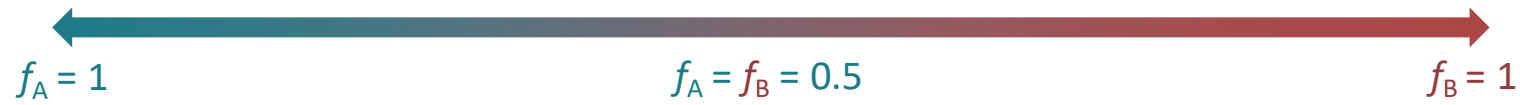
Prof. Dr. Guillaume Delaittre

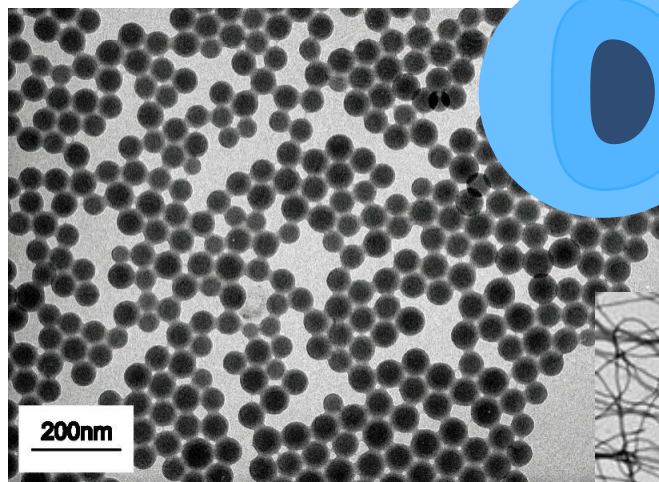
Chair of Macromolecular Chemistry

Summer Semester 2026

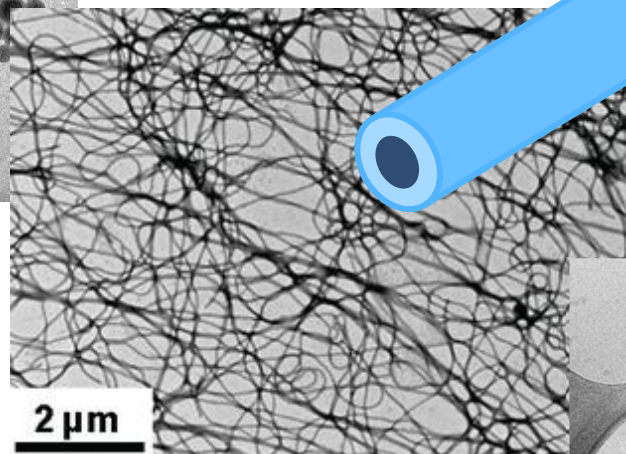
Trailer

A-Block B-Block

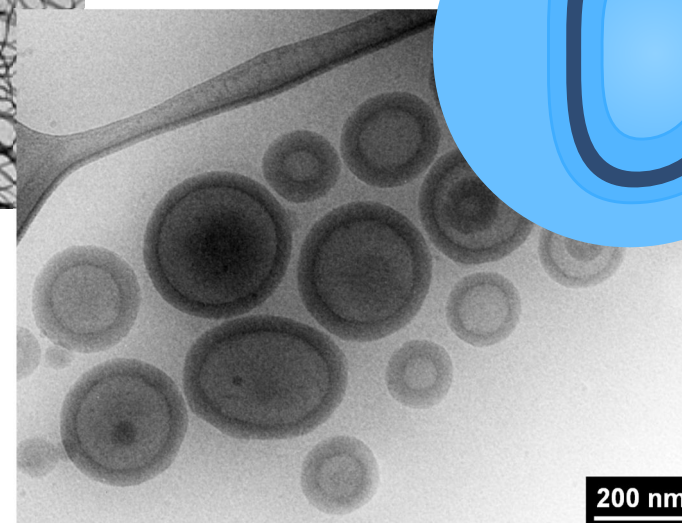




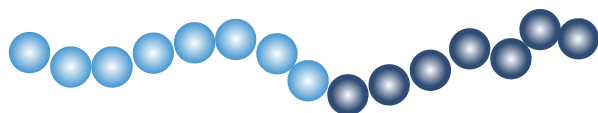
Delaittre et al.
Chem. Commun. **2005**, 614

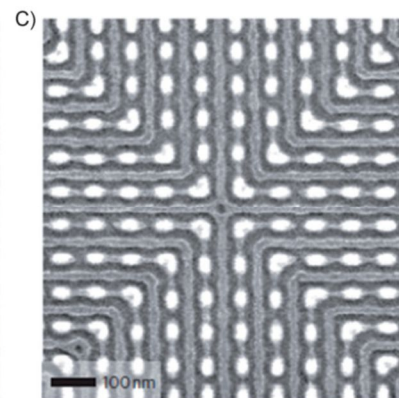
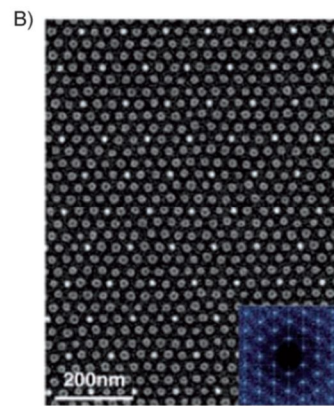
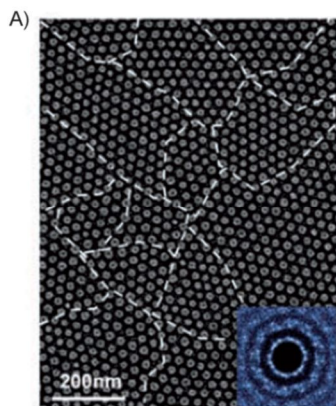
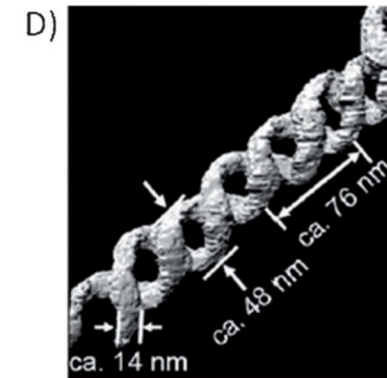
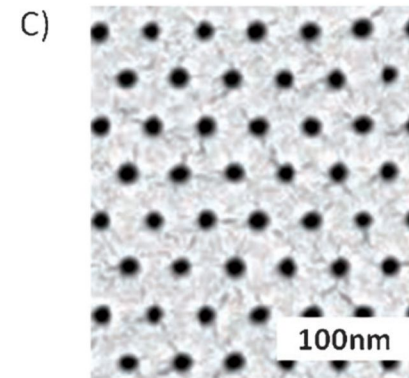
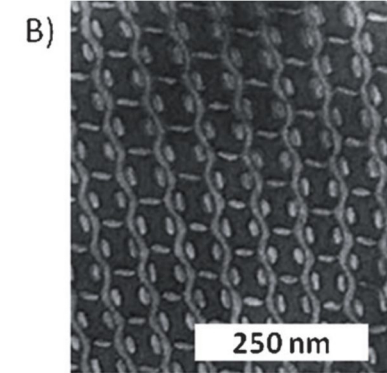
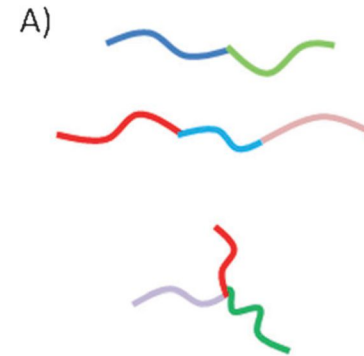
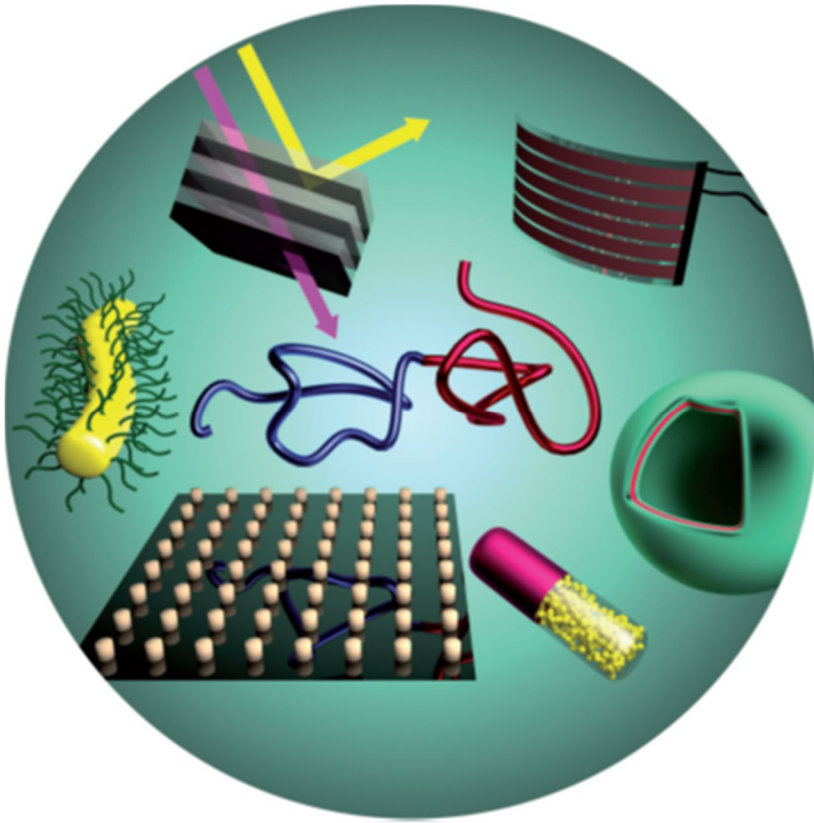


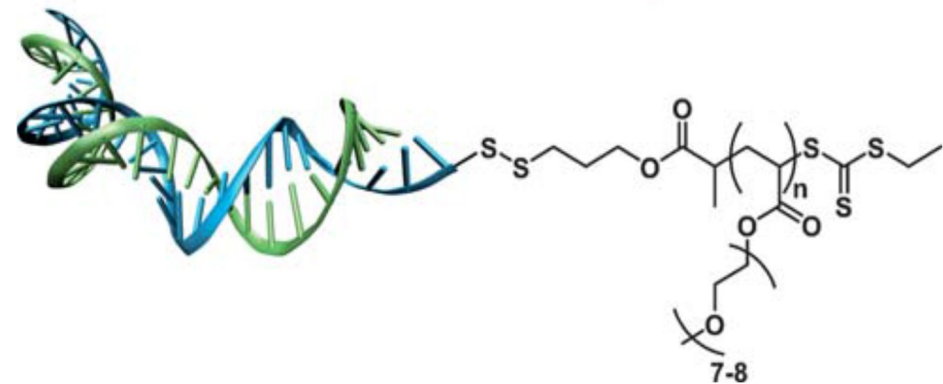
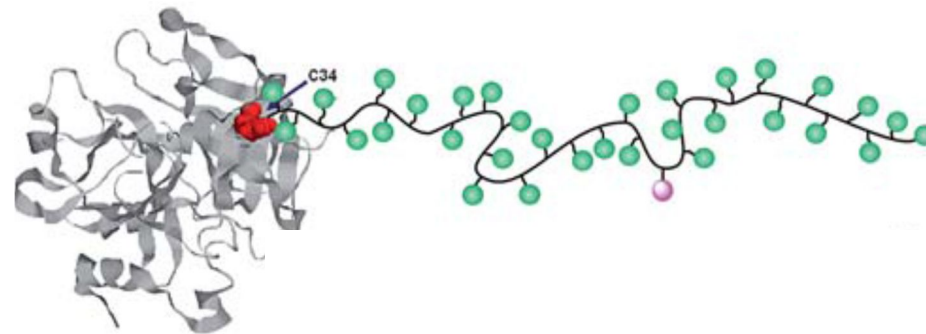
ACS Macro Lett. **2012**, 1, 47

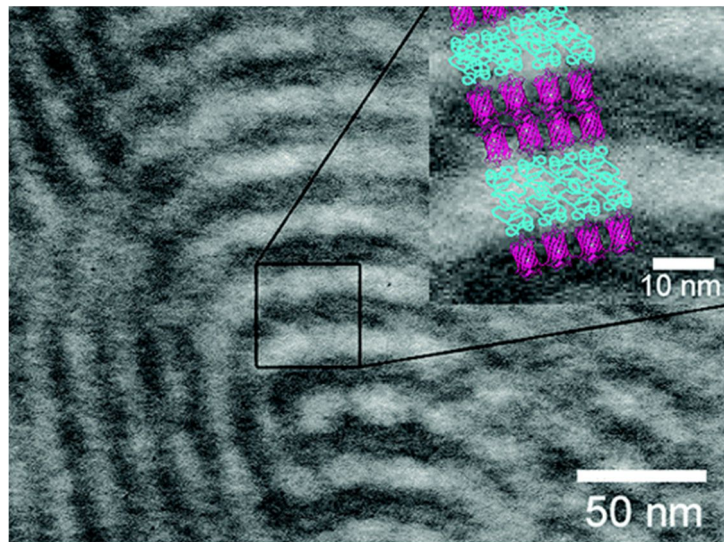


Delaittre et al.
Chem. Commun. **2009**, 2887

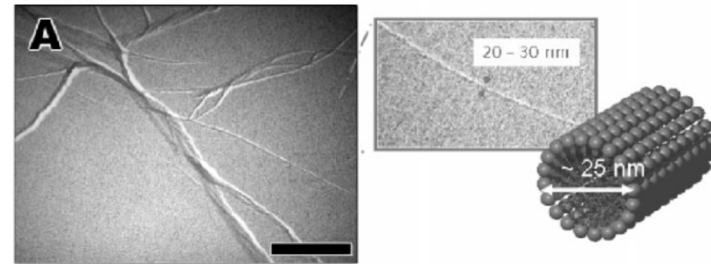




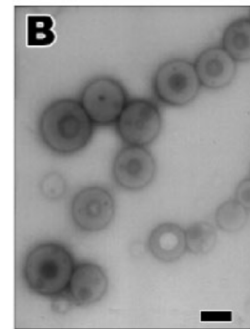




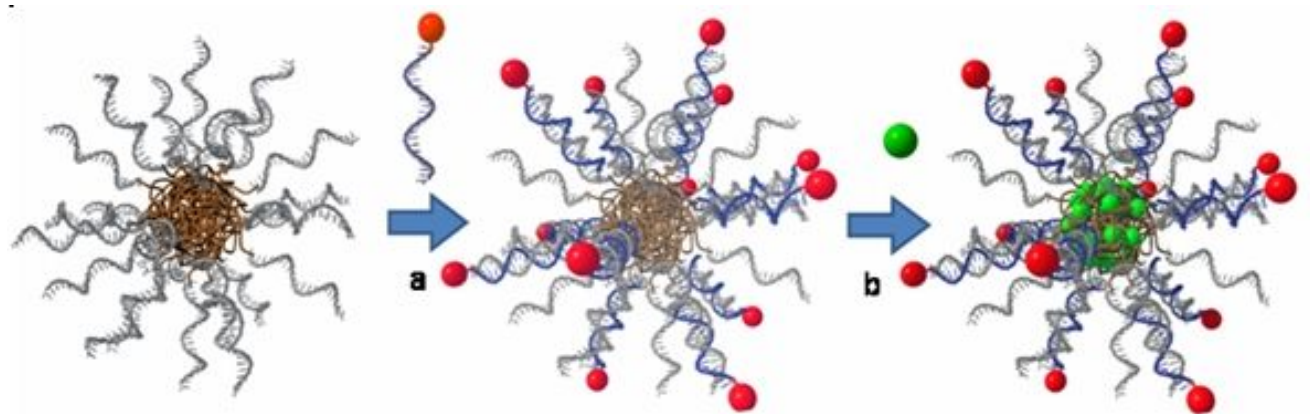
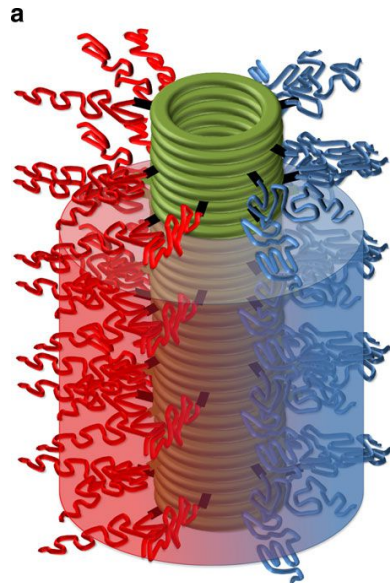
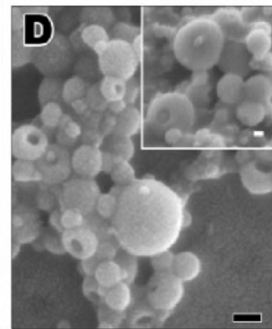
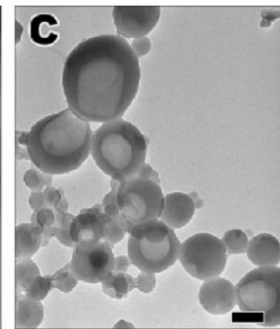
CalB-PS

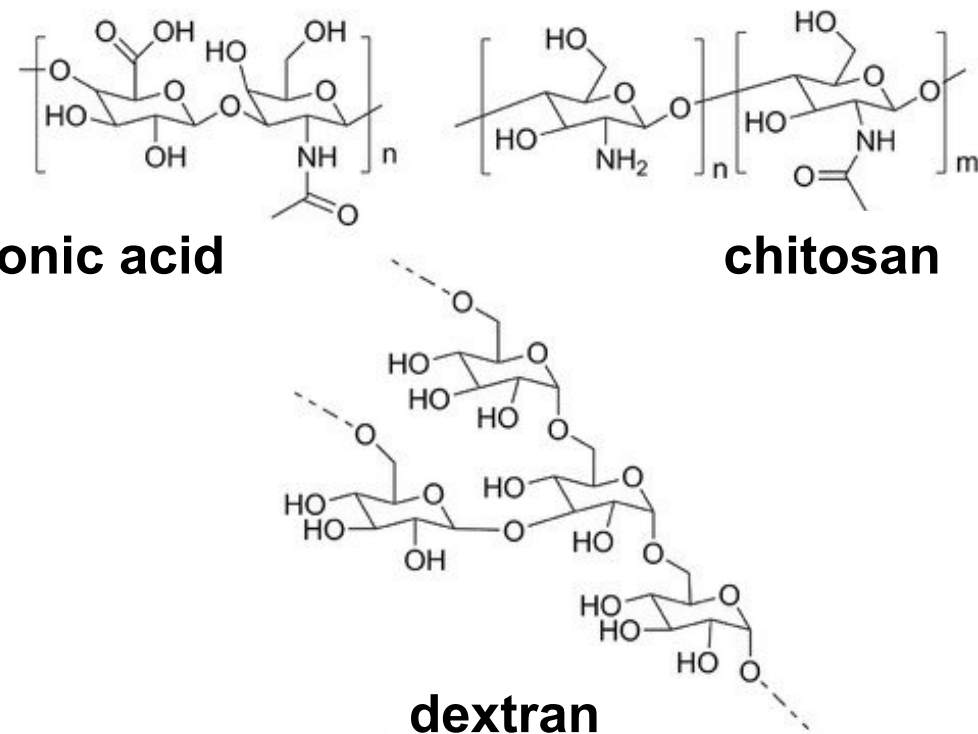
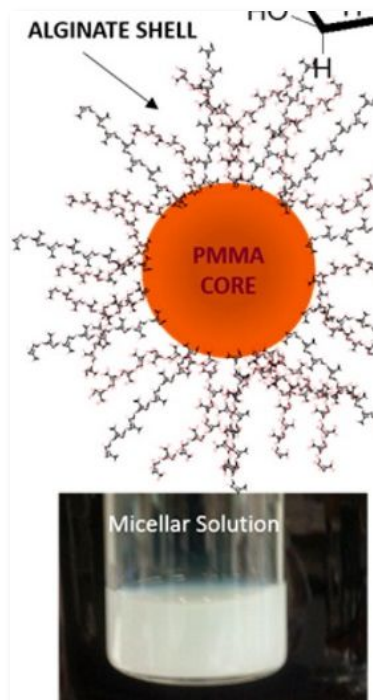
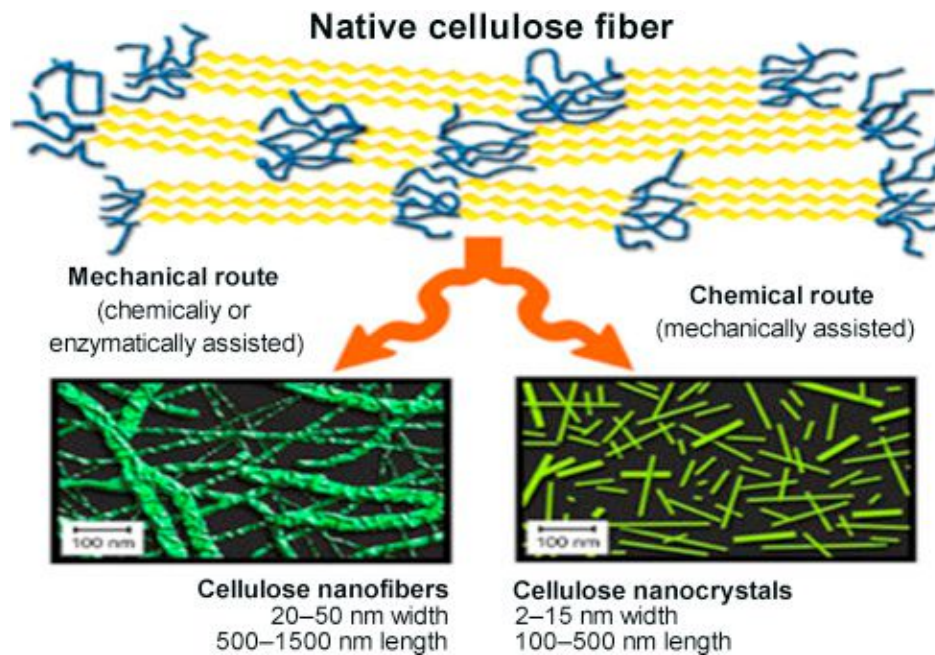


HRP-PS



Mb-PS





Lecture format

- Block Lecture – 7 x 90 min:
 - Monday, June 1 – 15:30
 - Wednesday, June 3 – 08:30
 - Monday, June 8 – 15:30
 - Tuesday, June 9 – 15:30
 - Thursday, June 11 – 15:30
 - Monday, June 15 – 15:30
 - Wednesday, June 17 – 08:30
- English

Marking format

- BSc and MSc (MC-QM) – oral exam, together with your direct supervisor (including practical work)
- MSc (AMC) – oral exam together with Dr. Tabatabai (Vorlesung GTS)

Synthetic Block Copolymers

- **Main types of block copolymers**
- **Synthetic methods**
 - + spontaneous block copolymerization
 - + sequential monomer addition
 - + coupling of segments
- **Molecular characterization of block copolymers**
- **Self-assembly**
 - + in the solid state
 - origin
 - characterization
 - applications
 - + in solution
 - theoretical aspects and types of nanostructures
 - preparation
 - characterization
 - applications

Biomolecule-Polymer Conjugates

- **Basic concepts**
- **Polymer–protein conjugates**
 - + general aspects
 - + synthesis
 - + characterization
 - + applications
- **Other examples**

(Purely) Synthetic Block Copolymers

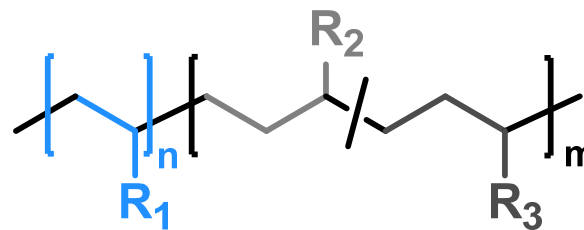
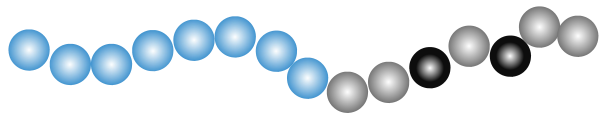
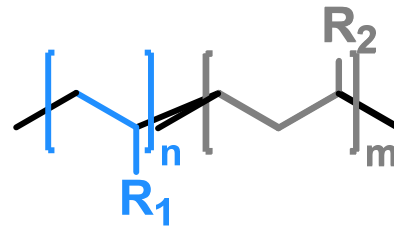
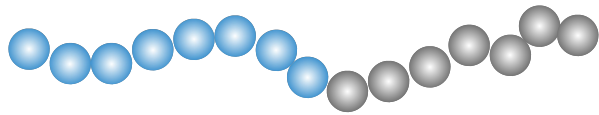
- General Concepts -

(Purely) Synthetic Block Copolymers

- General Concepts -

What is a Block Copolymer?

Definition / Notation

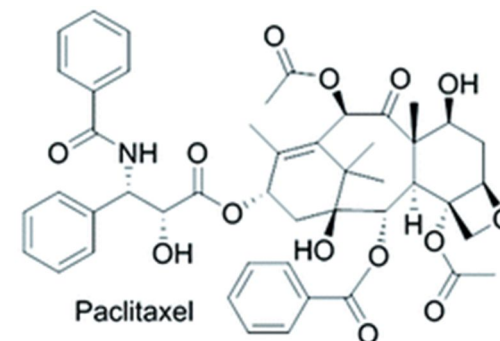
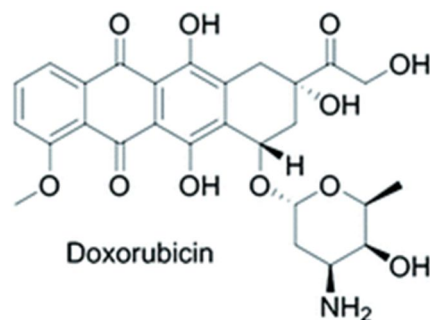
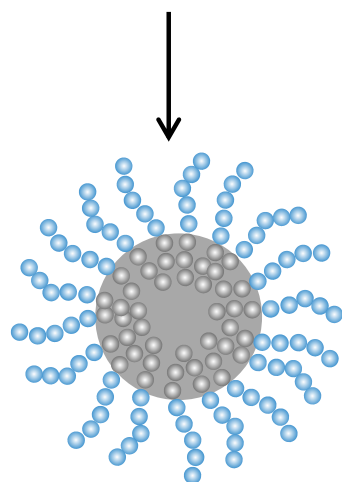
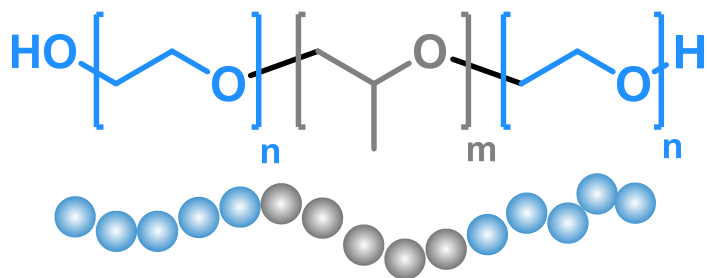


Why Block Copolymers?

Bringing distinct properties in one macromolecule = at (almost) the same location

Example of Pluronics (poloxamers):

- one **hydrophobic** block (*e.g.*, adhesive to surfaces)
- the other ones bringing **hydrophilicity**

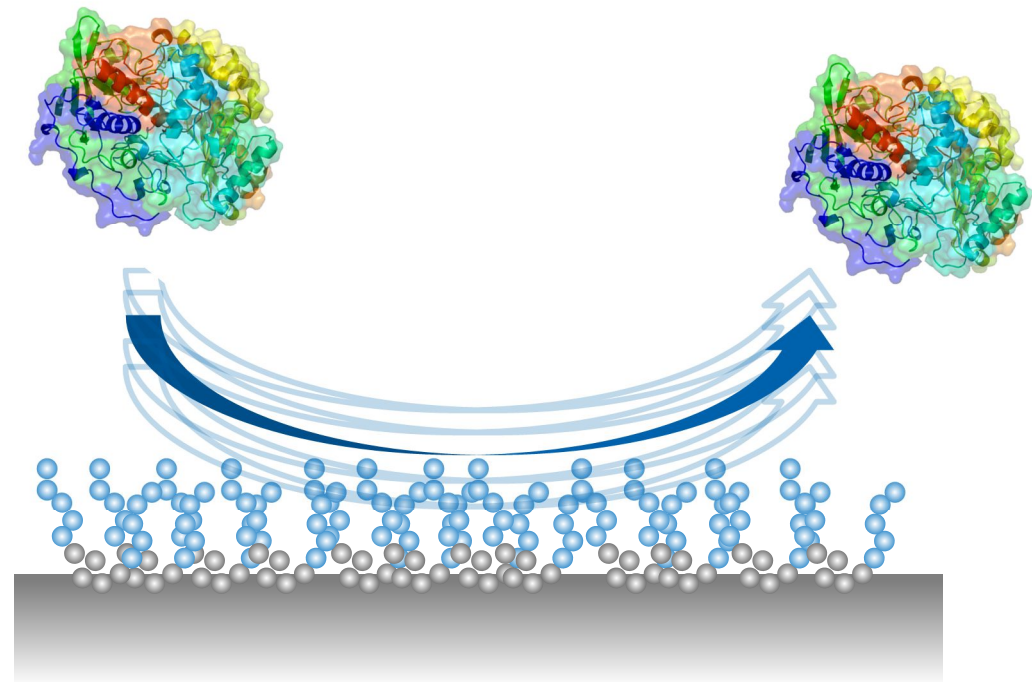
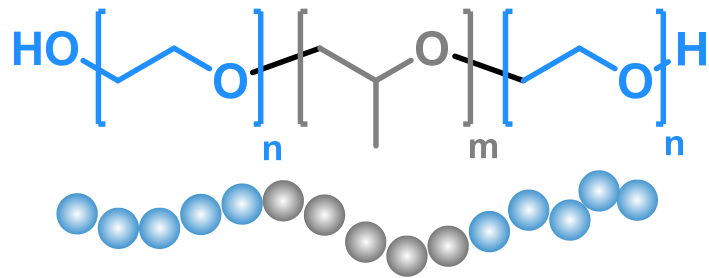


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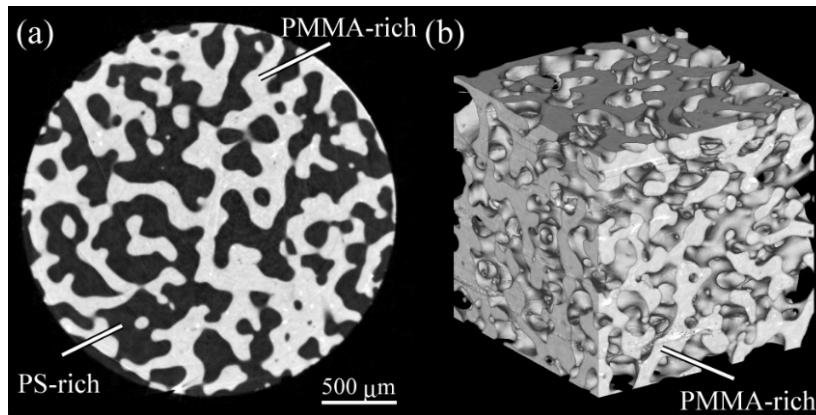
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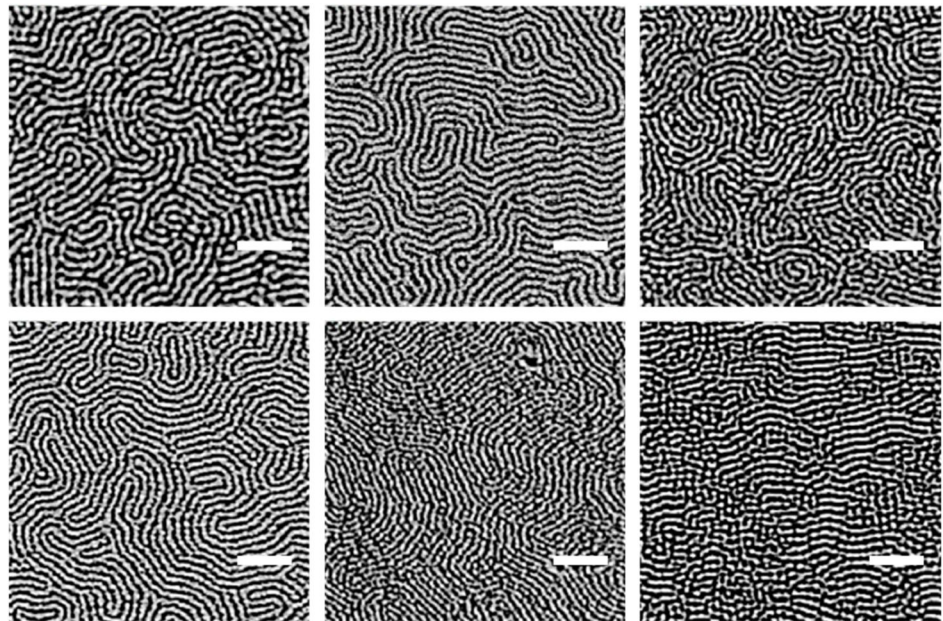
Why Block Copolymers?

Most polymers are incompatible: they do NOT mix in the melt.

Imagine what happens when two incompatible polymers are connected to each other...



*X-ray phase tomographic image
of a PS/PMMA blend*

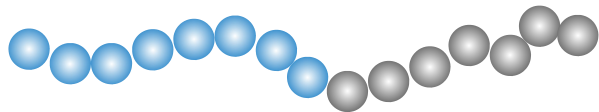


*Atomic force micrograph of PS-b-PMMA
Scale bars 200 nm*

Types of Block Copolymers

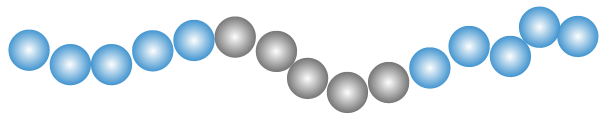
According to the Chemical Definition

Linear Diblock Copolymers

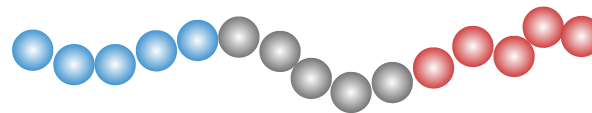


Linear Triblock Copolymers

Symmetrical



Asymmetrical

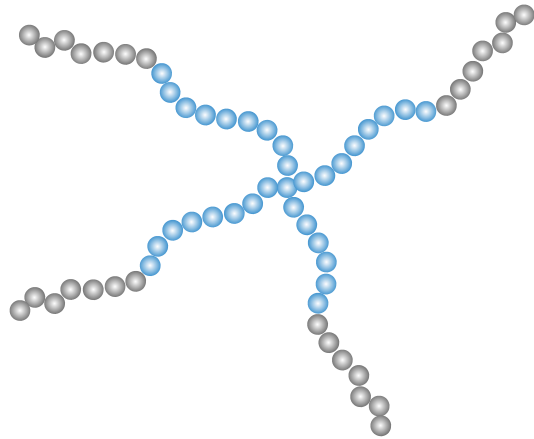


Types of Block Copolymers

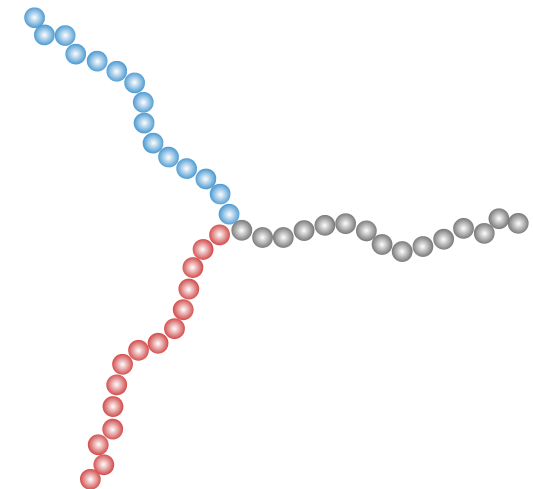
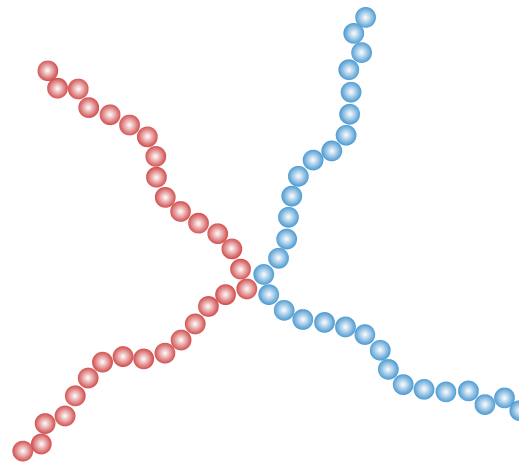
According to the Chemical Definition

Star Block Copolymers

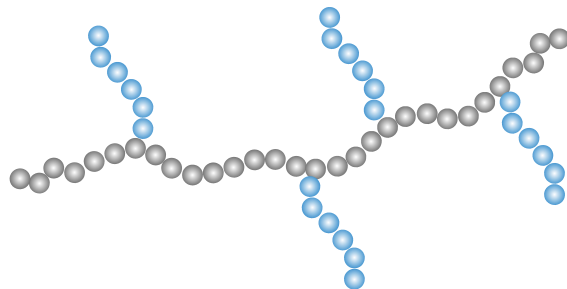
All arms identical



Mikto-arm star copolymers ($\mu\kappa\tau\acute{o}\varsigma$ = mixed)



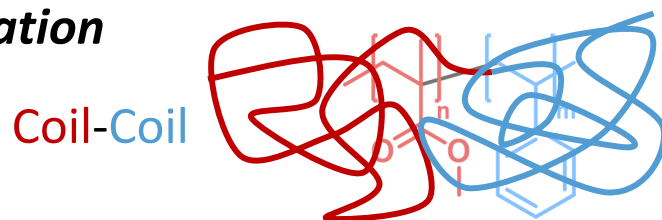
Graft Copolymers



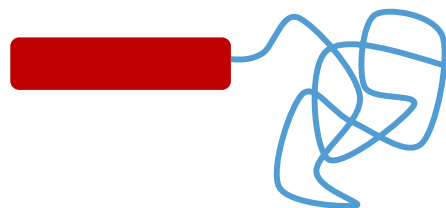
Types of Block Copolymers

According to Physical Properties

Conformation



Rod-Coil



Rod-Rod

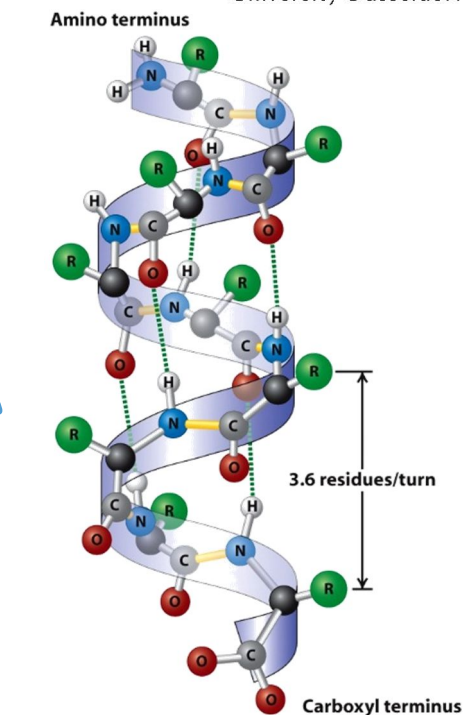
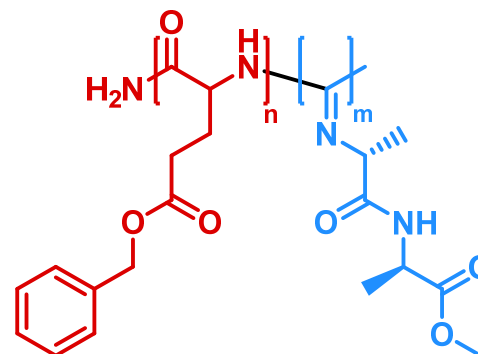
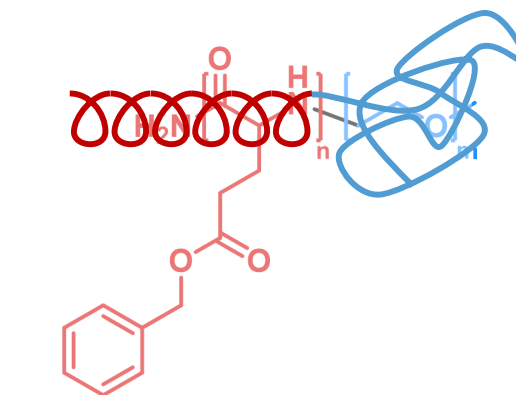


Figure 3-4
Molecular Cell Biology, Sixth Edition
© 2008 W. H. Freeman and Company

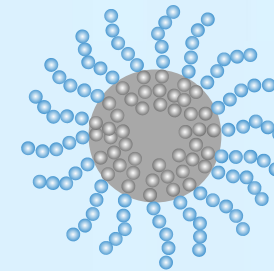
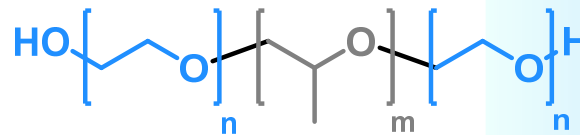
Chem. Commun. **2009**, 2709
Angew. Chem. Int. Ed. **2005**, 44, 4349
Polym. Chem. **2011**, 2, 300

Types of Block Copolymers

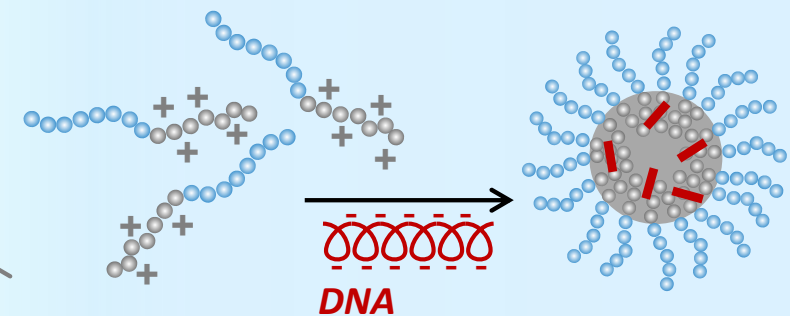
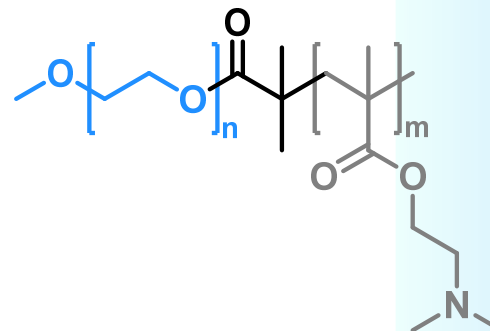
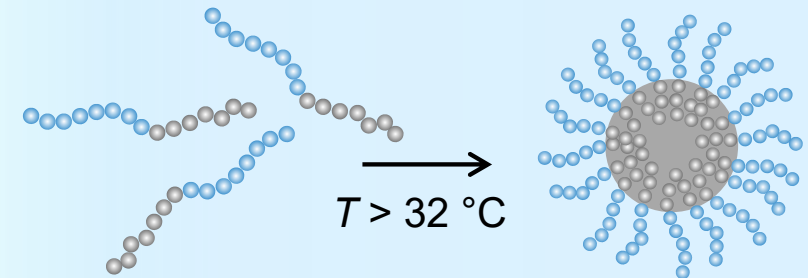
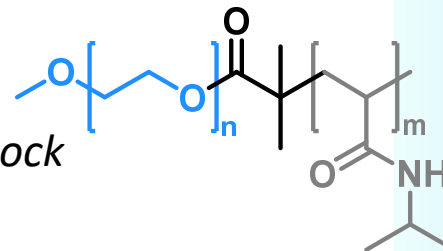
According to Physical Properties

Hydrophilicity

Amphiphilic



Double Hydrophilic
often with a responsive block



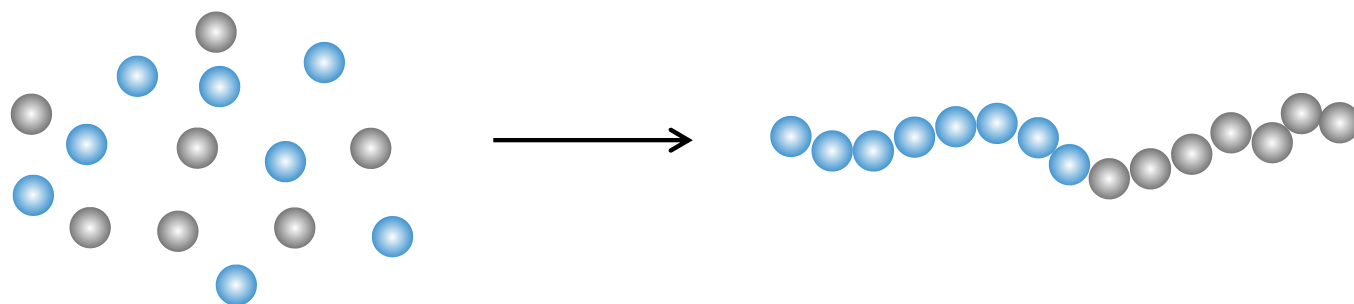
In water

(Purely) Synthetic Block Copolymers ***- Synthetic Strategies -***

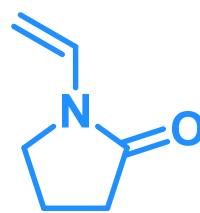
Synthesis of Linear Block Copolymers

Spontaneous Block Copolymerization (“One Shot”)

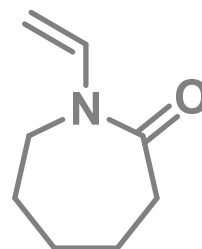
Favorable reactivity ratios: $r_1 > 1$ and $r_2 > 1$



“very rare”... had to go back to the 50s/60s for radical copolymerization!



$$r_{\text{NVP}} = 2.8$$

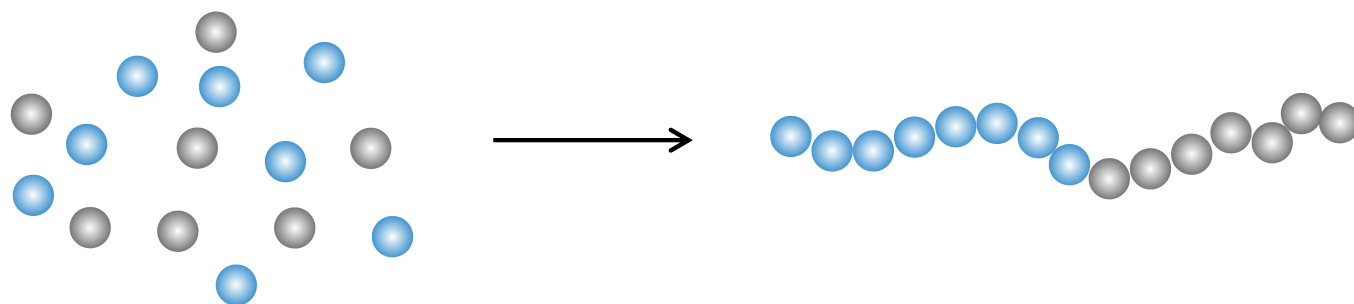


$$r_{\text{NVCL}} = 1.7$$

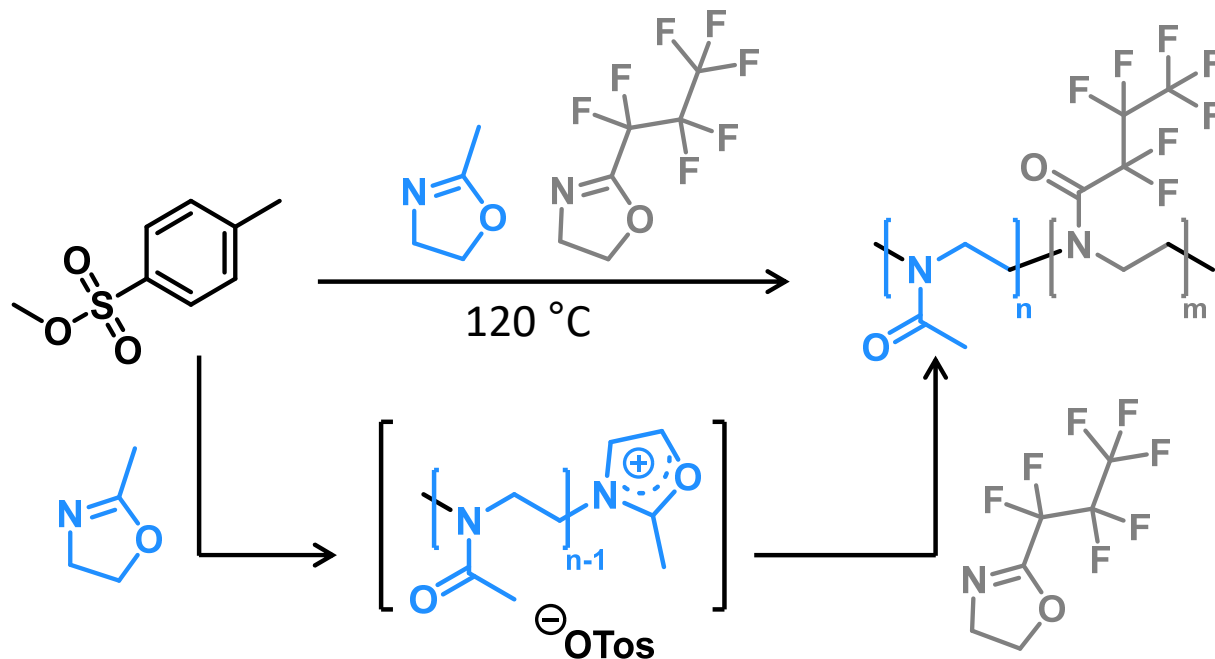
Synthesis of Linear Block Copolymers

Spontaneous Block Copolymerization ("One Shot")

Favorable reactivity ratios: $r_1 > 1$ and $r_2 > 1$



Polyoxazolines with perfluoroalkyl substituents

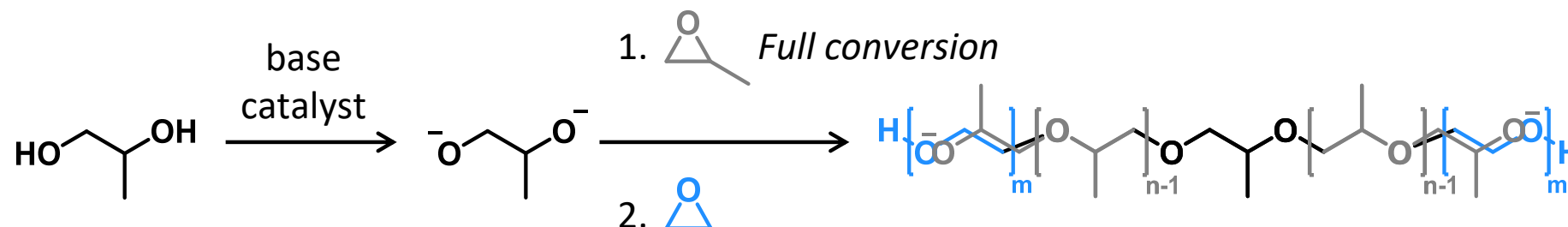


Prog. Polym. Sci. **1996**, 21, 151

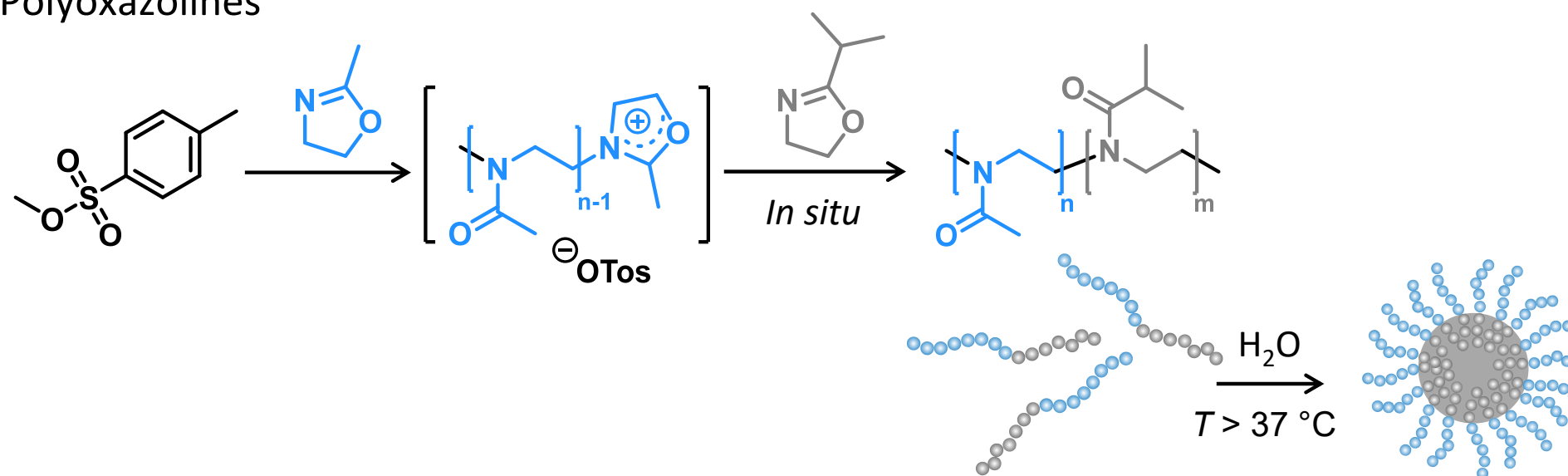
Synthesis of Linear Block Copolymers

Sequential Addition of Monomers ("One-Pot Two-Stage")

Pluronics

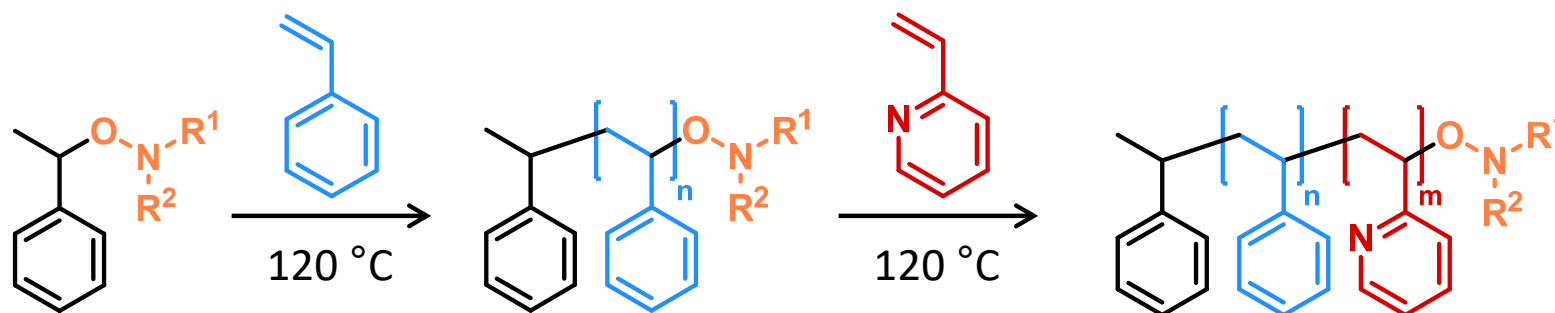


Polyoxazolines



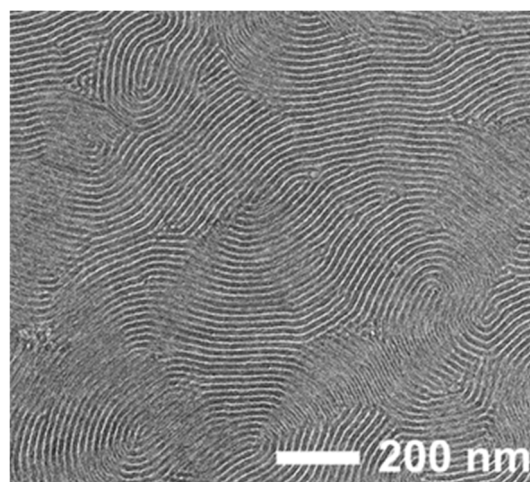
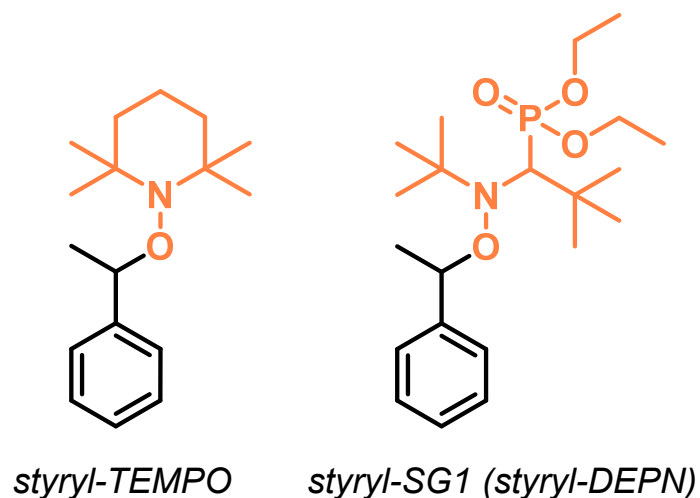
Synthesis of Linear Block Copolymers

Macroinitiator / Macromolecular Transfer Agent

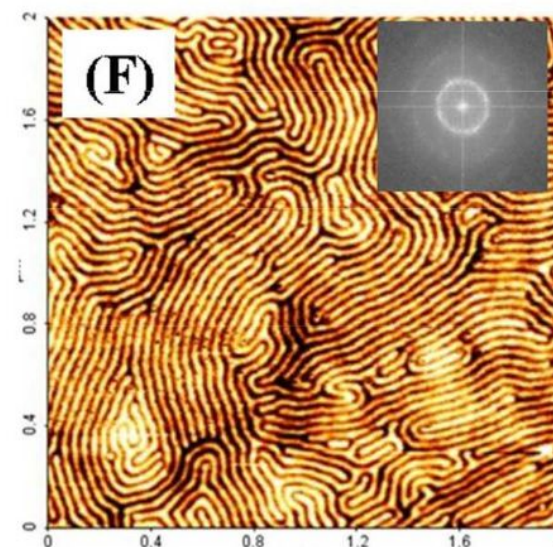


conversion < 70–80 %

Purification by precipitation in methanol



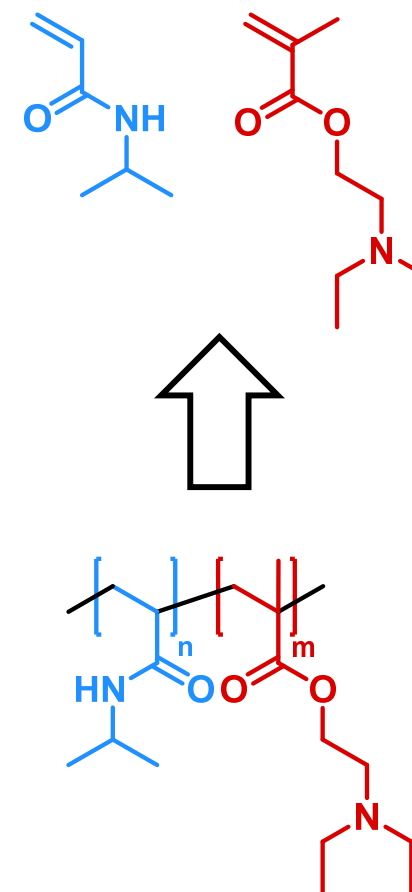
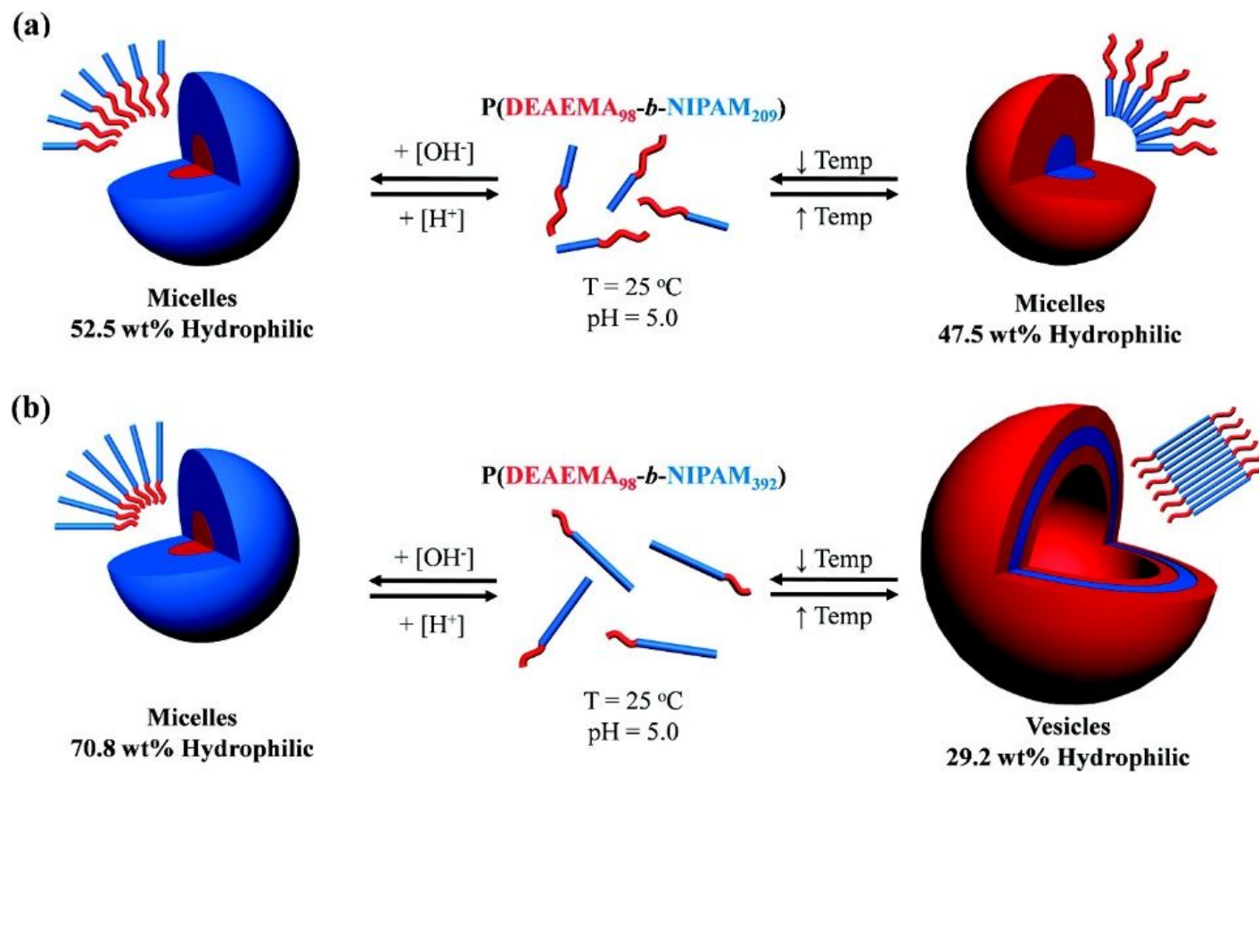
Adv. Mater **2005**, 17, 2591



Soft Matter **2016**
DOI: 10.1039/C6SM00815A

Synthesis of Linear Block Copolymers

Macroinitiator / Macromolecular Transfer Agent



Synthesis of Linear Block Copolymers



Graeme
Moad



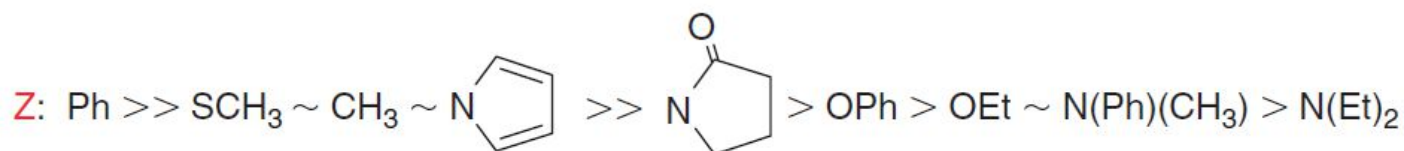
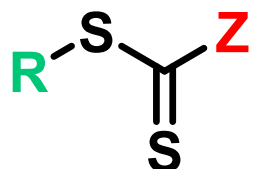
Ezio
Rizzardo



San H.
Tang



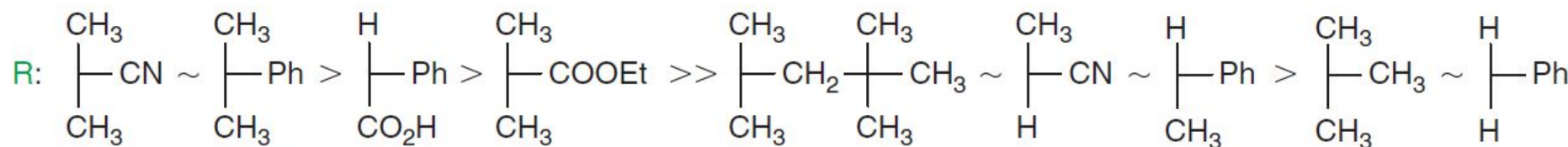
Macroinitiator / Macromolecular Transfer Agent



← MMA →

← VAc →

← S, MA, AM, AN →



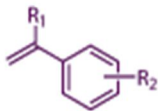
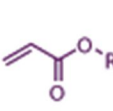
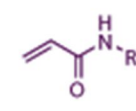
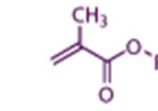
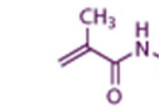
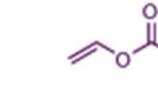
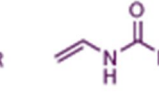
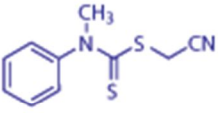
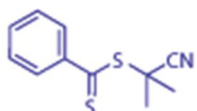
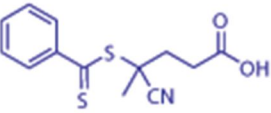
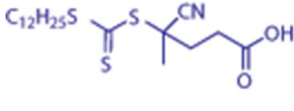
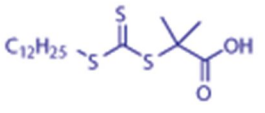
← MMA →

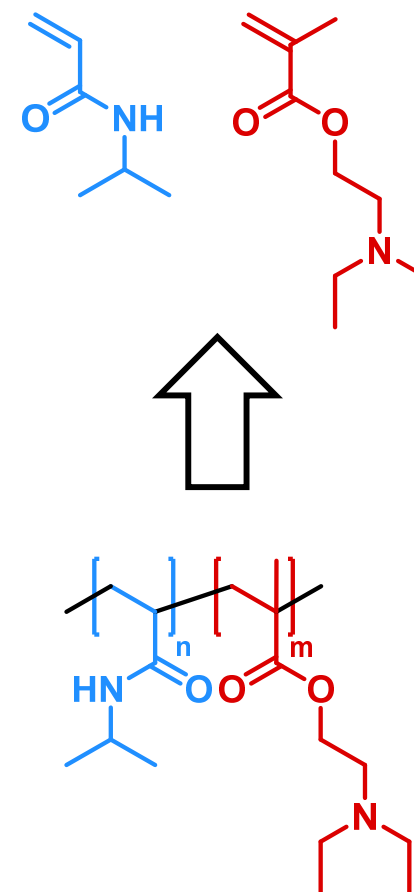
← S, MA, AM, AN →

← VAc →

Synthesis of Linear Block Copolymers

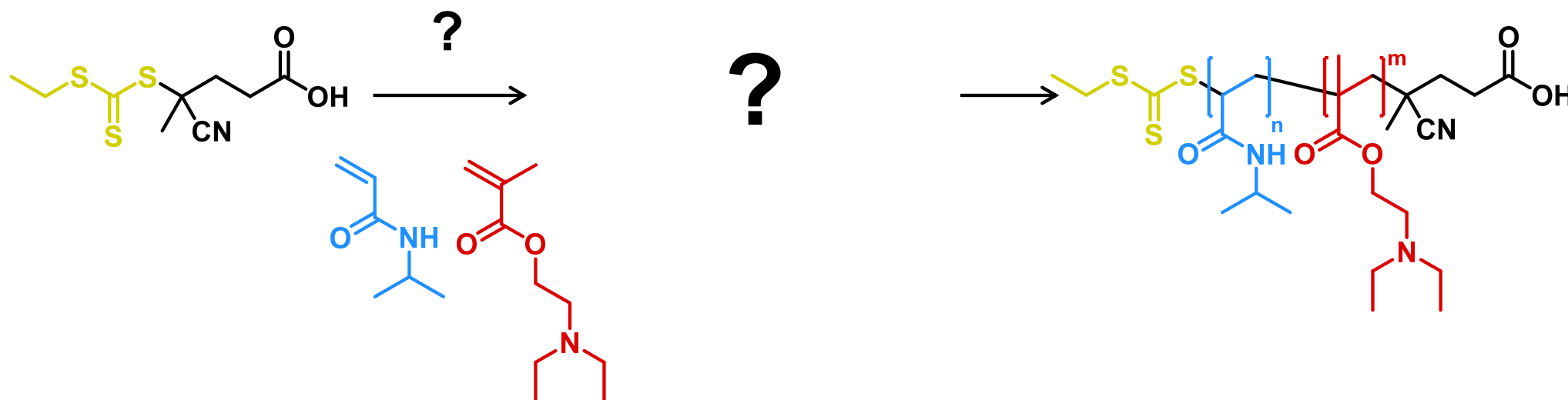
Macroinitiator / Macromolecular Transfer Agent

							
	styrenes	acrylates	acrylamides	methacrylates	methacrylamides	vinyl esters	vinyl amides
	—	—	—	—	—	+++	+++
	+++	+++	+++	—	—	—	—
	++	+	—	+++	+++	—	—
	++	+	+	+++	+++	—	—
	+++	++	++	+++	+++	—	—
	+++	++	++	+++	+++	—	—
	+++	+++	+++	+	+	—	—



Sigma-Aldrich Website

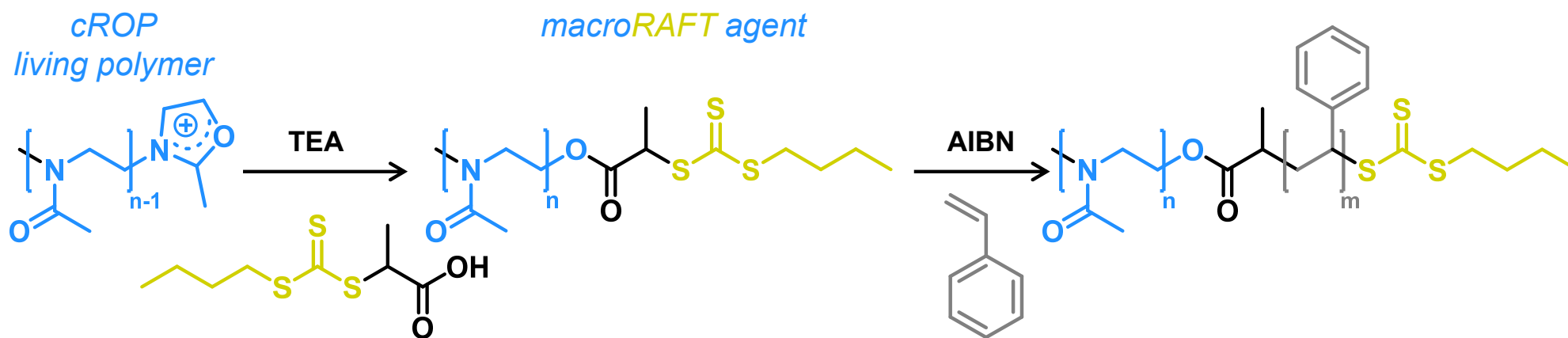
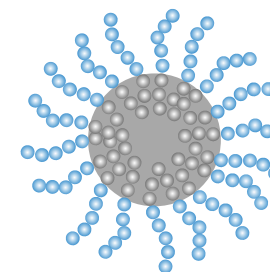
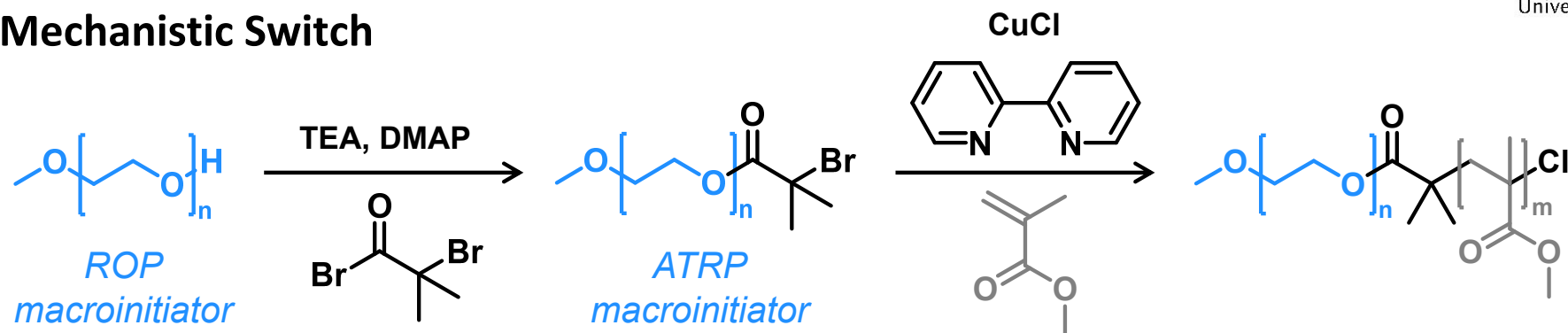
Macroinitiator / Macromolecular Transfer Agent



Which one goes first?

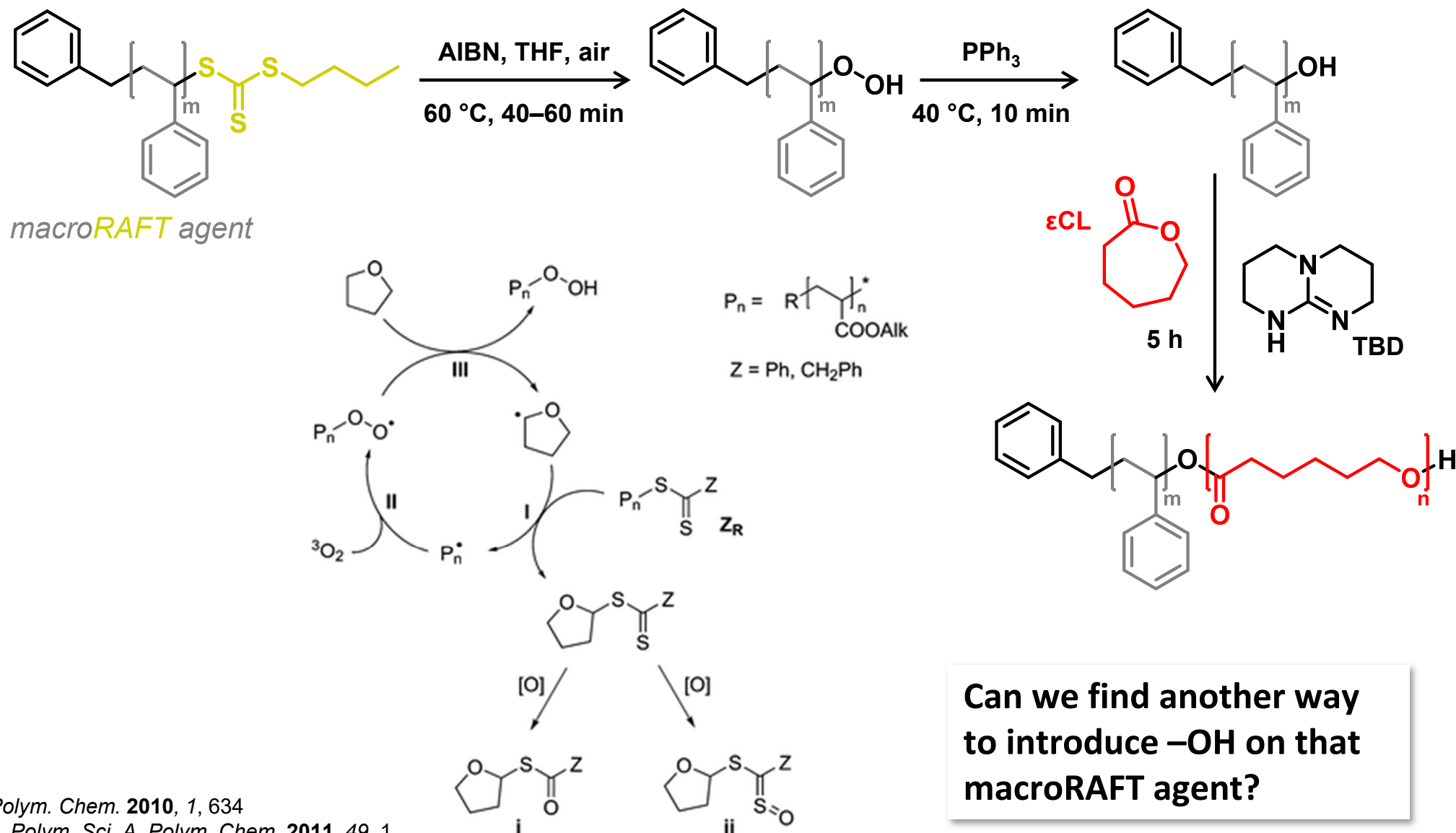
Synthesis of Linear Block Copolymers

Mechanistic Switch



ACS Macro Lett. 2012, 1, 776

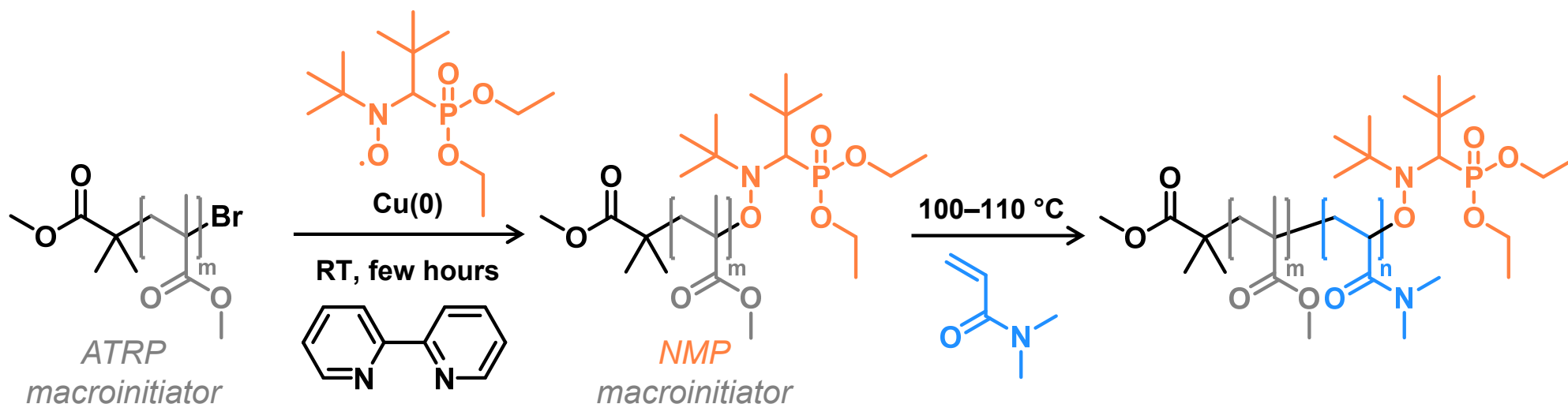
Mechanistic Switch



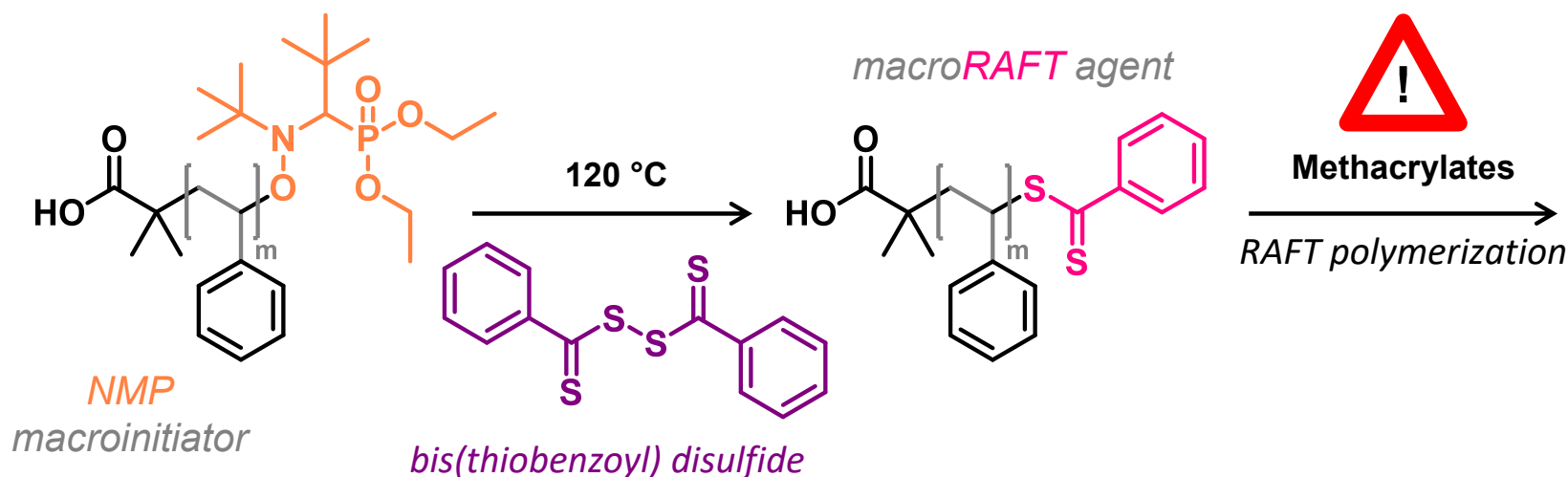
Can we find another way to introduce –OH on that macroRAFT agent?

Synthesis of Linear Block Copolymers

Pseudo-Mechanistic Switch



Polym. Chem. **2011**, 2, 1859

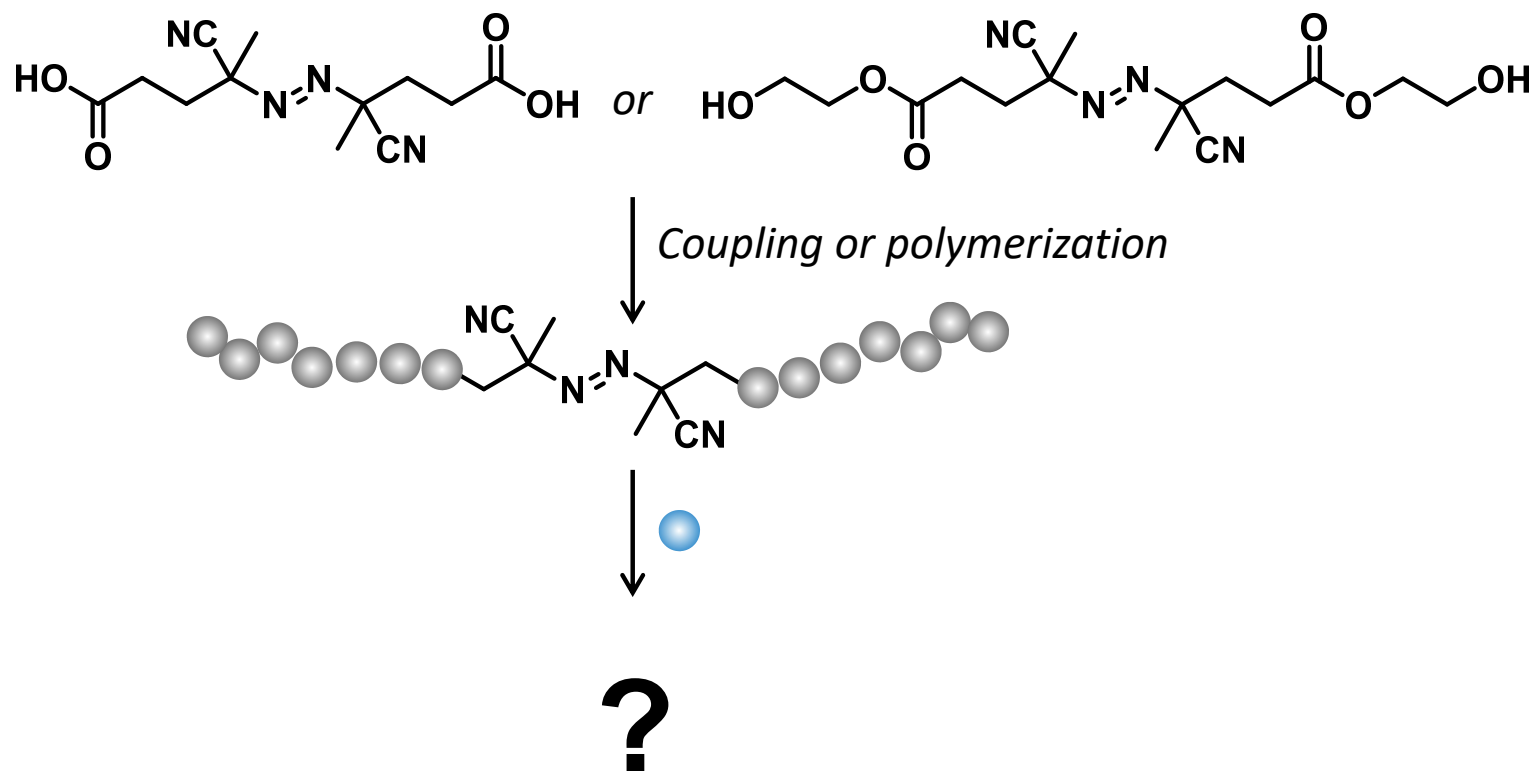


Macromol. Rapid Commun. **2012**, 33, 1310

Quick Note on Block Copolymers from Non-Controlled Polymerizations

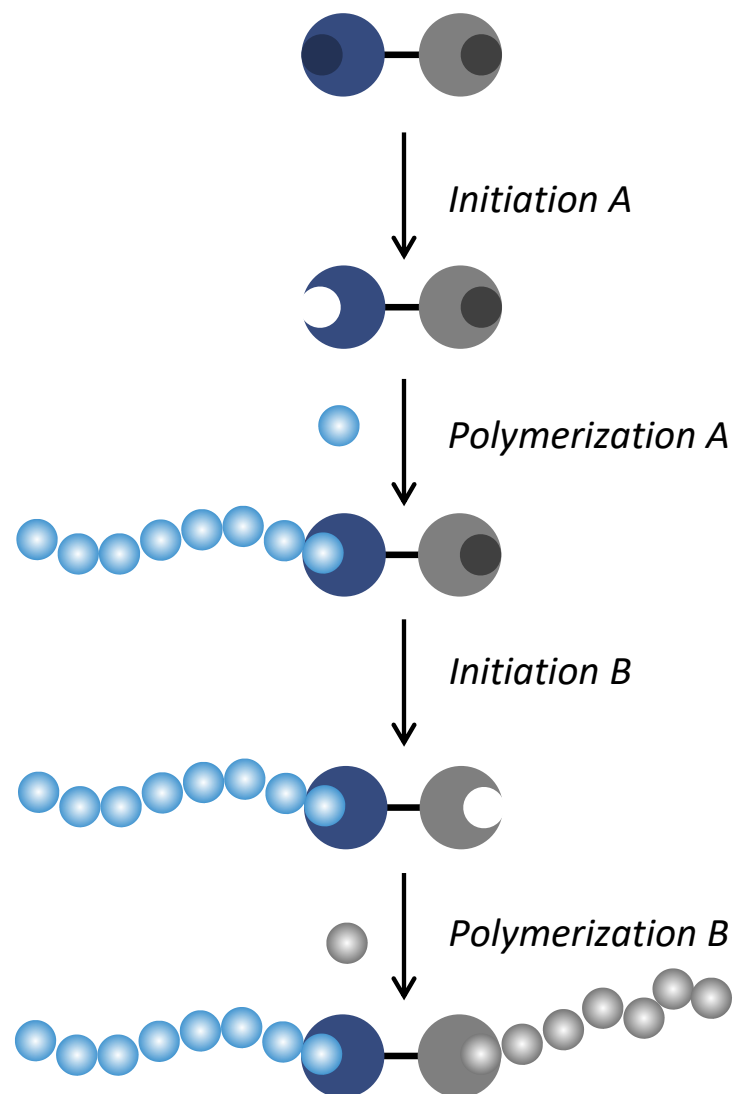
It is also possible to form block copolymers via non-controlled methods, however the definition of the final product is usually very poor!

Example of a radical diazo initiator with functional extremities



Synthesis of Linear Block Copolymers

Dual Initiator Route

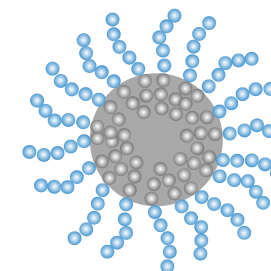
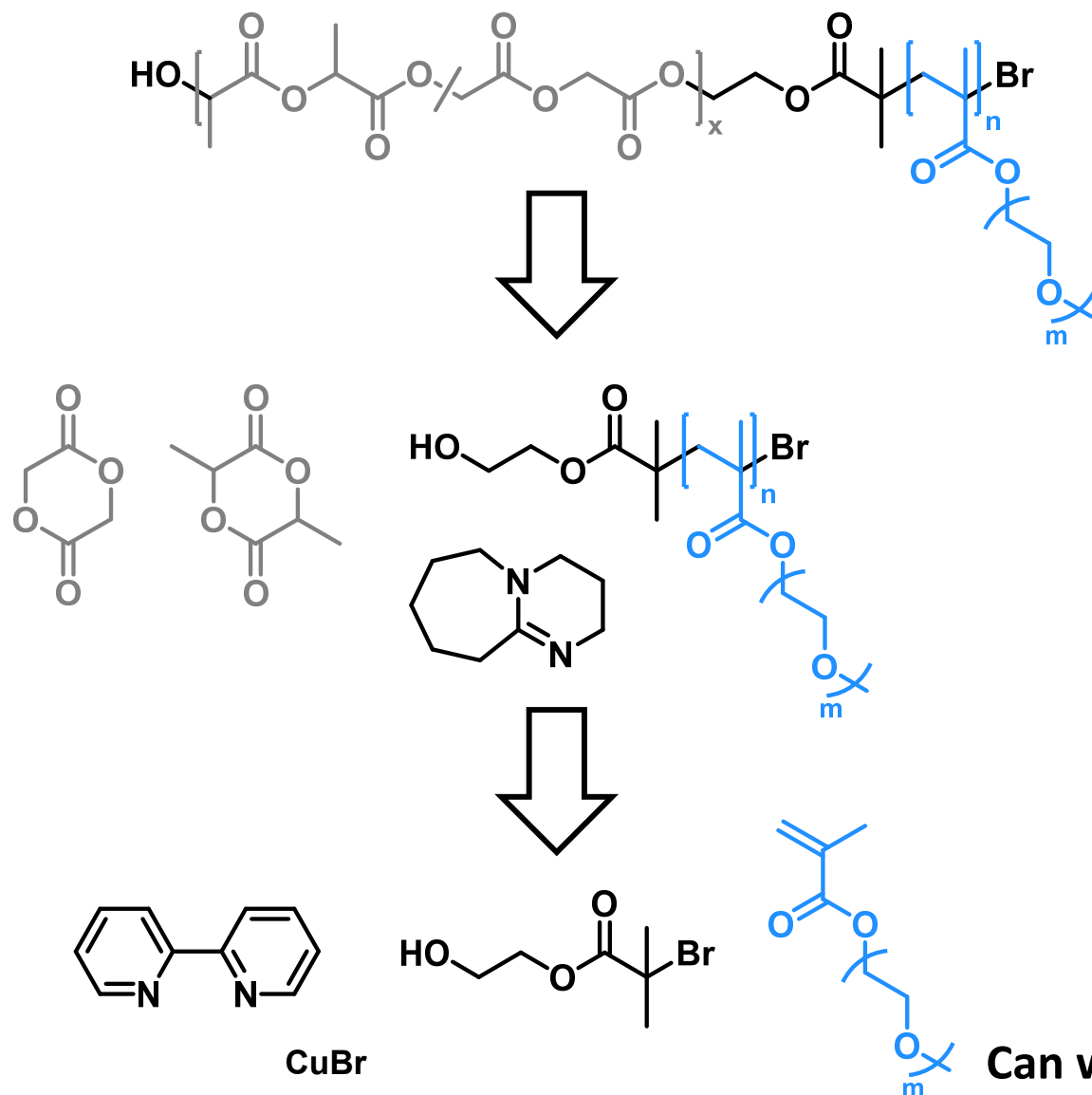


Prog. Polym. Sci. **2006**, 31, 671

Synthesis of Linear Block Copolymers

Dual Initiator Route – Purely distinct mechanisms

Sequential

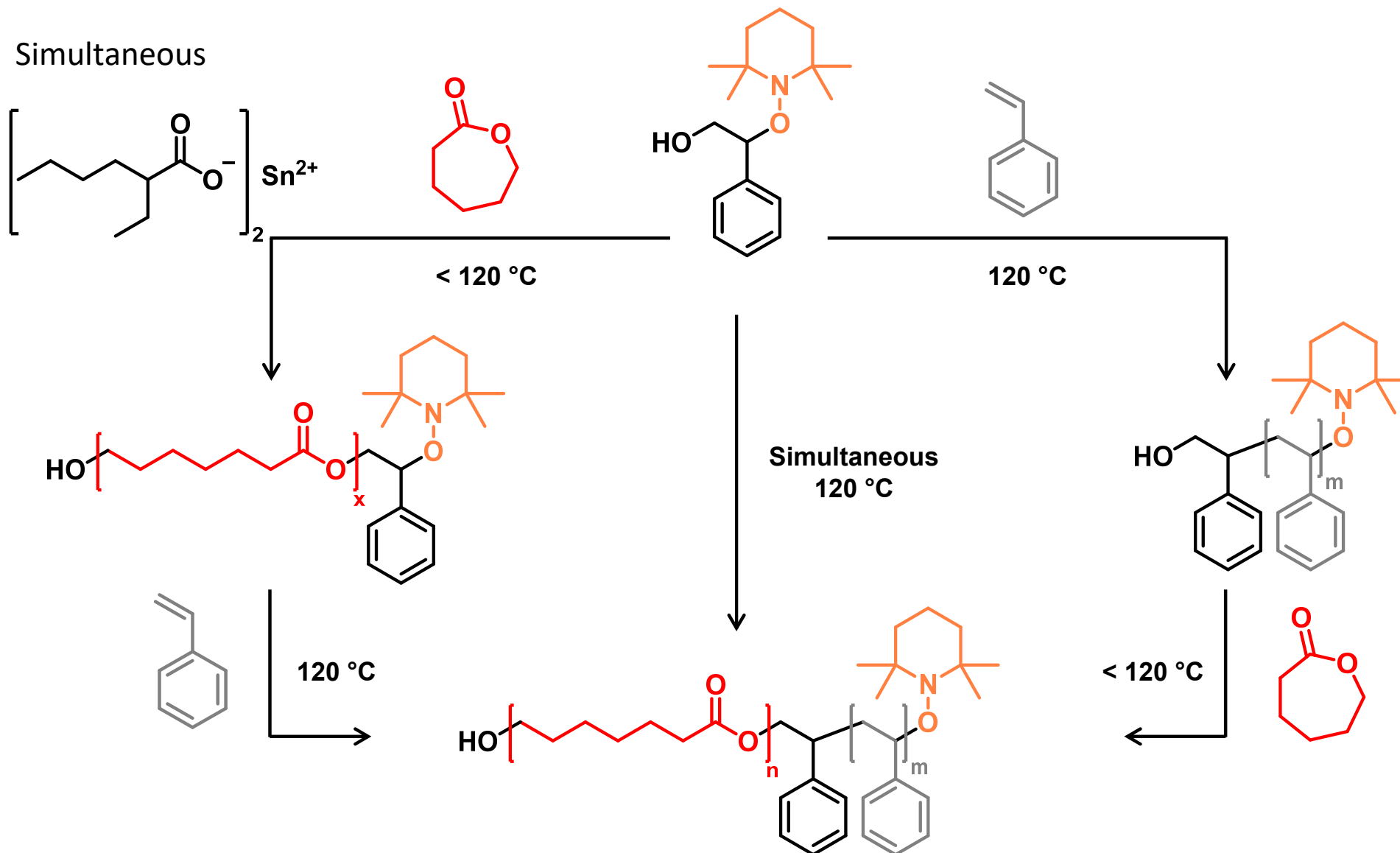


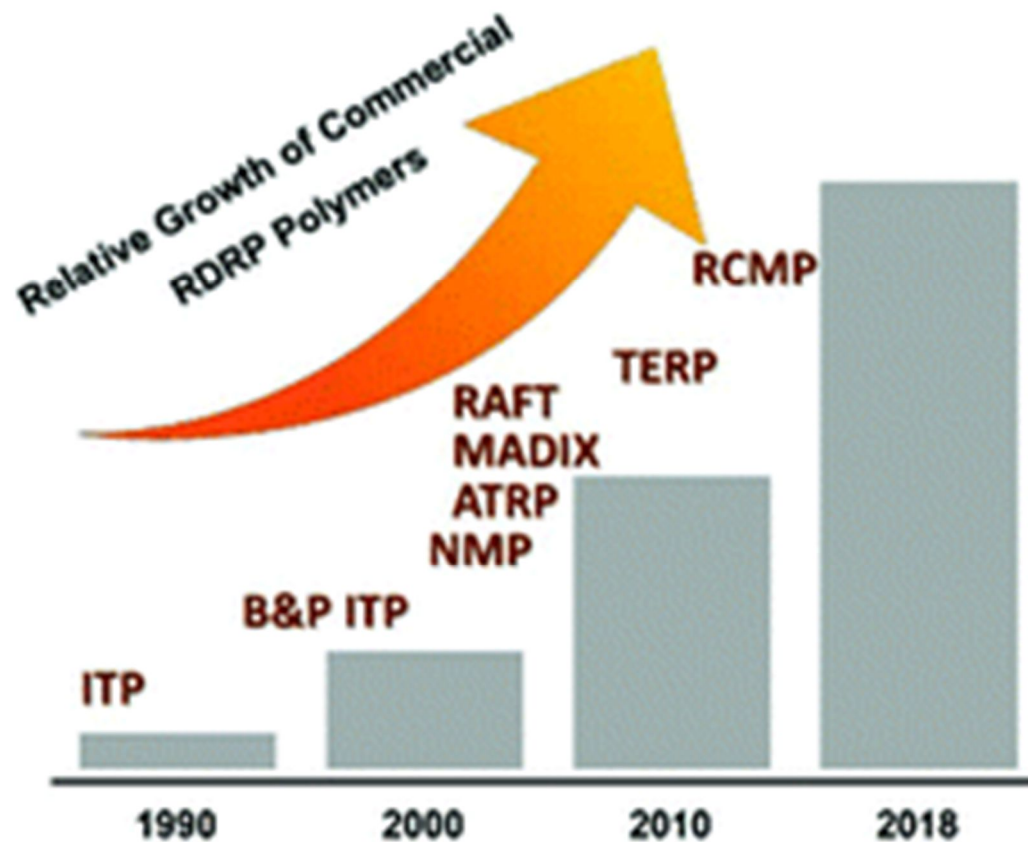
Can we reverse the order?

Synthesis of Linear Block Copolymers

Dual Initiator Route – Purely distinct mechanisms

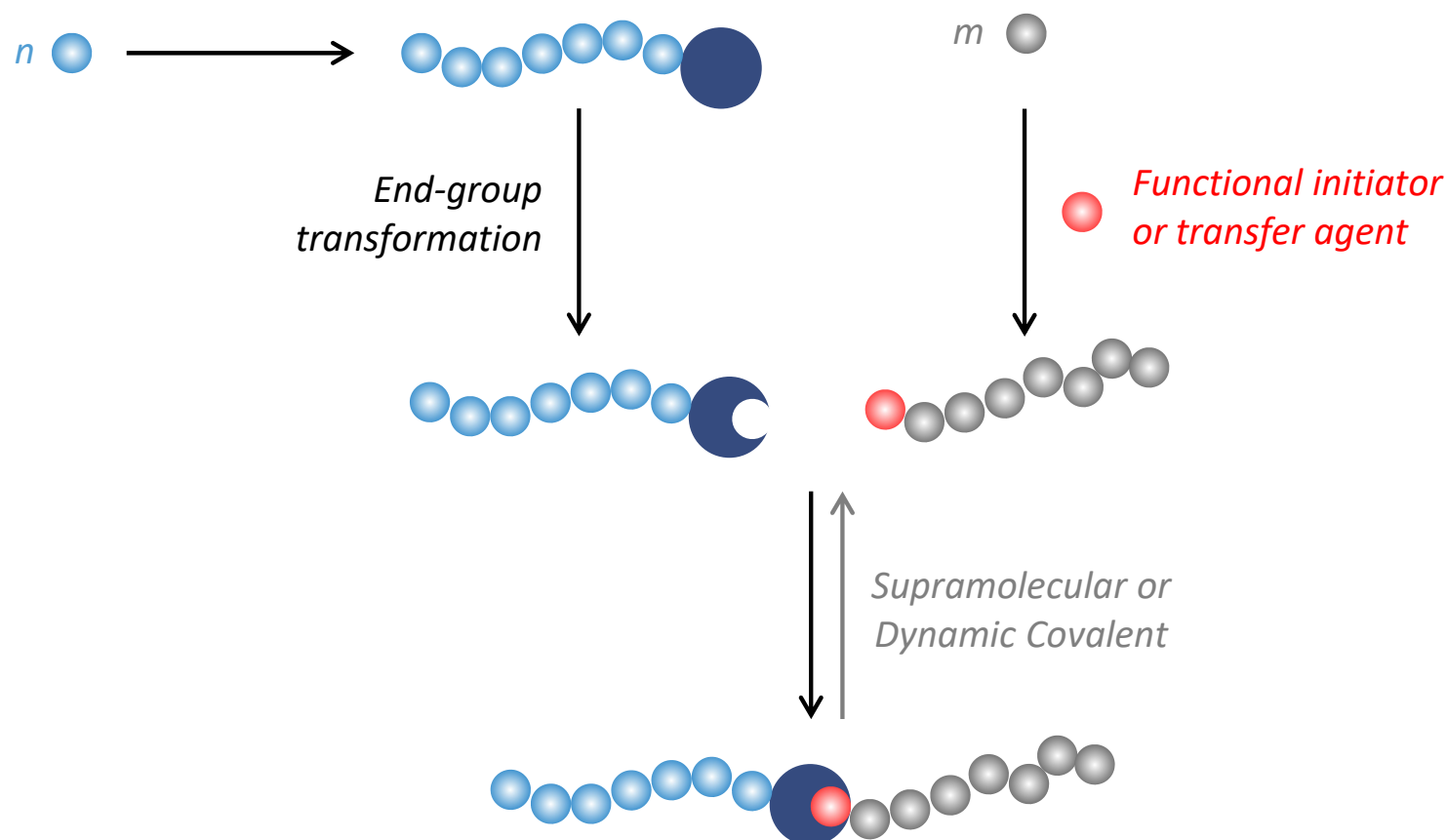
Simultaneous





M. Destarac *Polym. Chem.* **2018**, 9, 4947

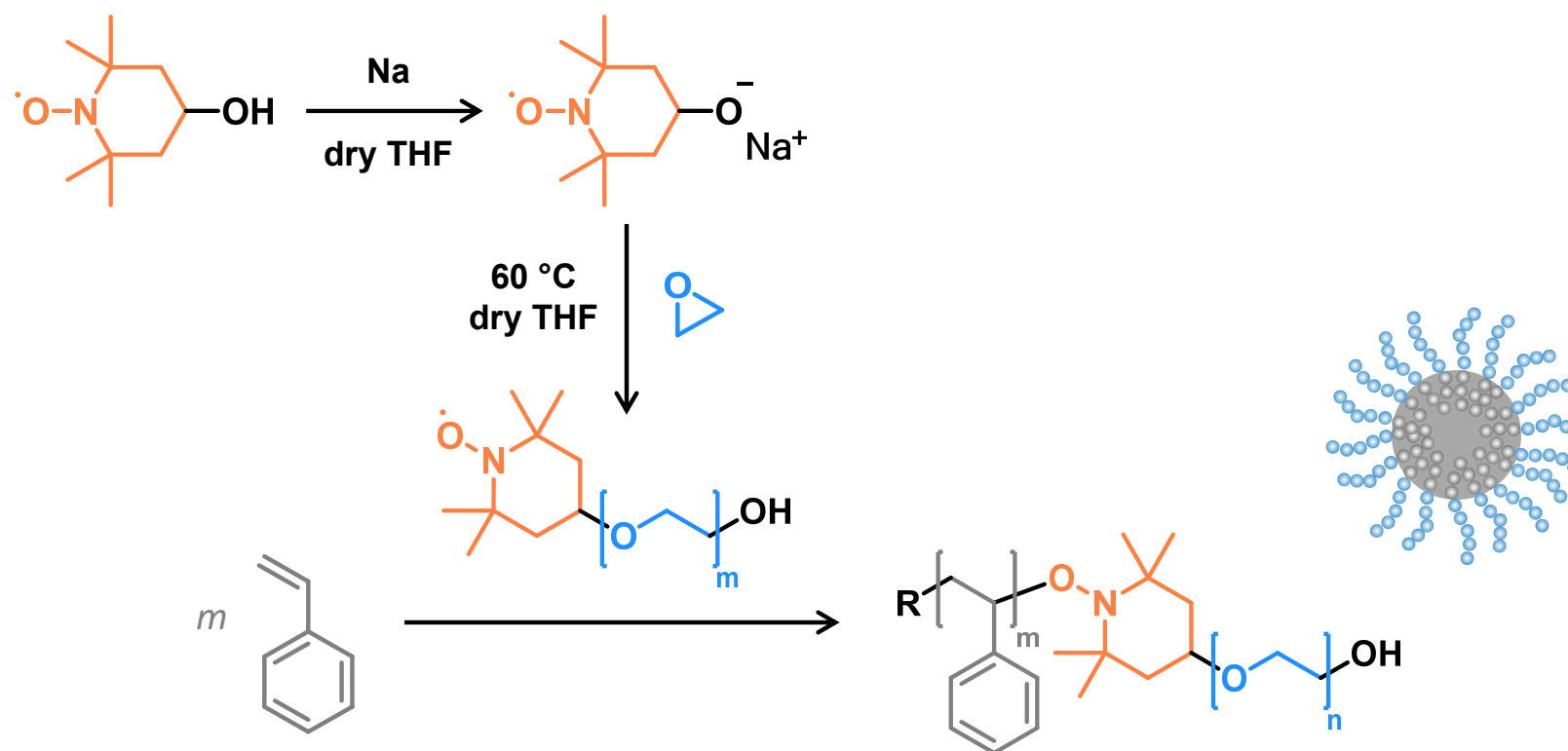
Coupling of Polymer Strands



Synthesis of Linear Block Copolymers

Coupling of Polymer Strands

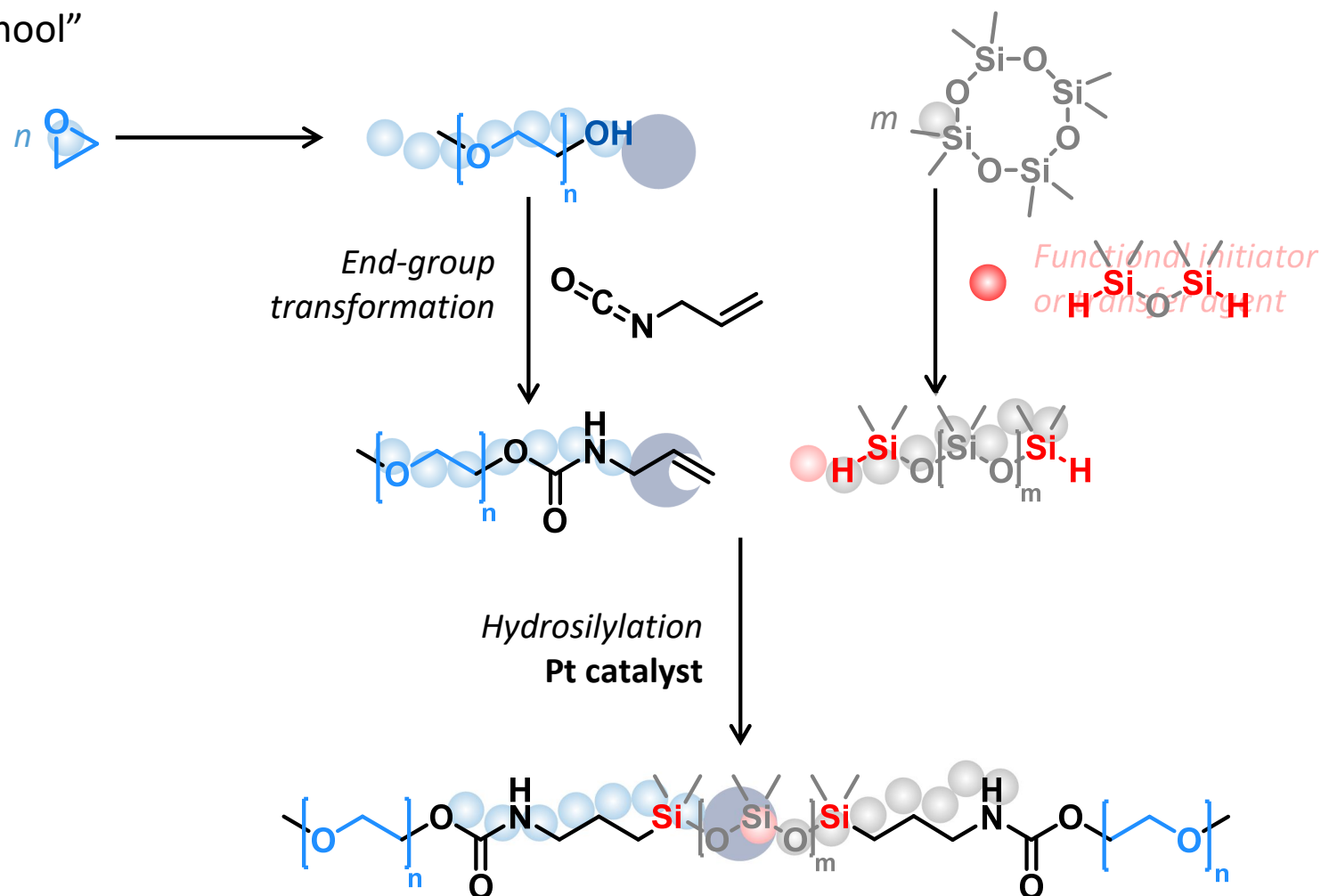
Special case of *initier*



Polymer **2001**, 42, 1361

Coupling of Polymer Strands

"Old School"

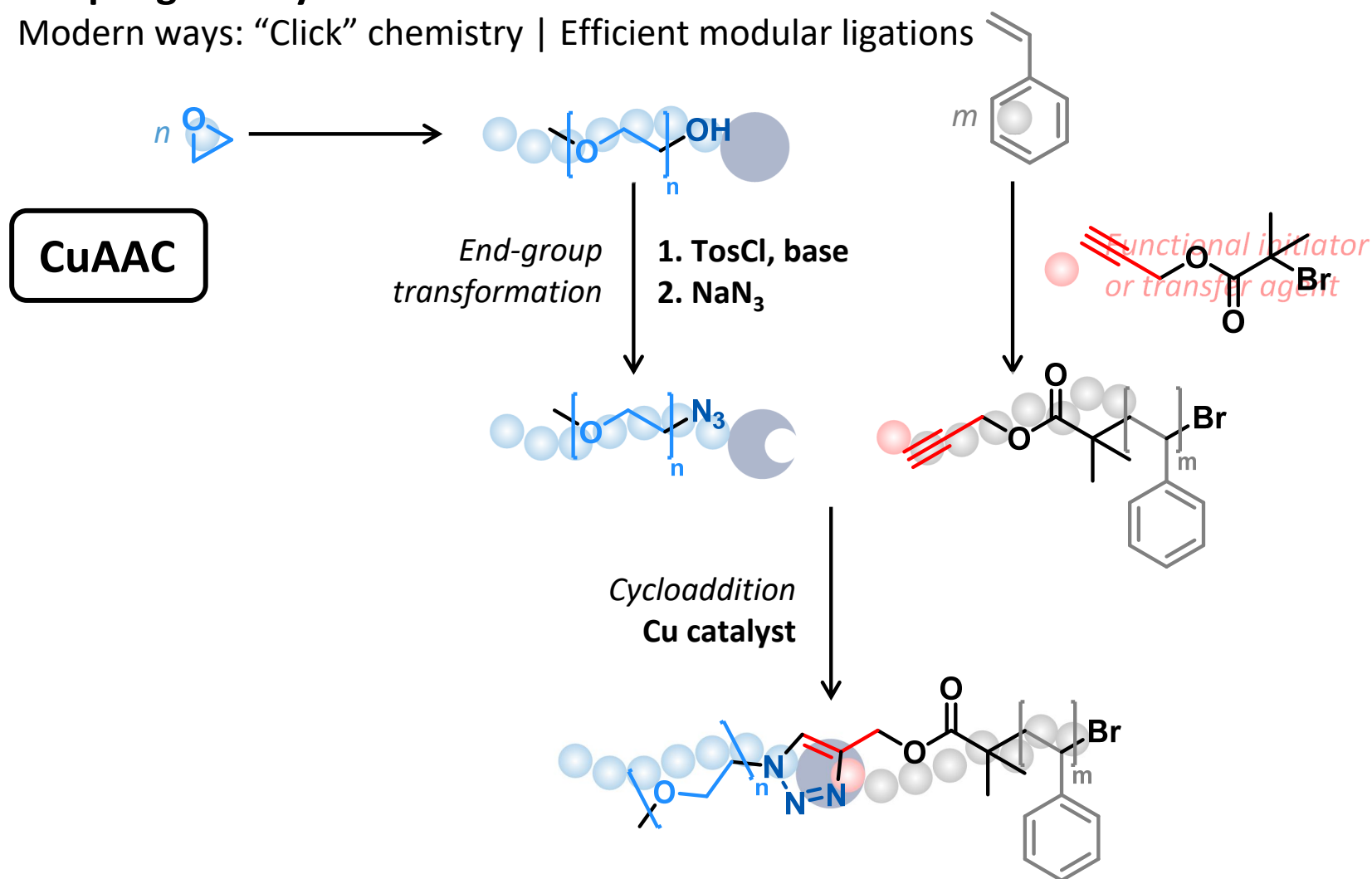


Makromol. Chem. **1985**, 186, 357

Synthesis of Linear Block Copolymers

Coupling of Polymer Strands

Modern ways: "Click" chemistry | Efficient modular ligations

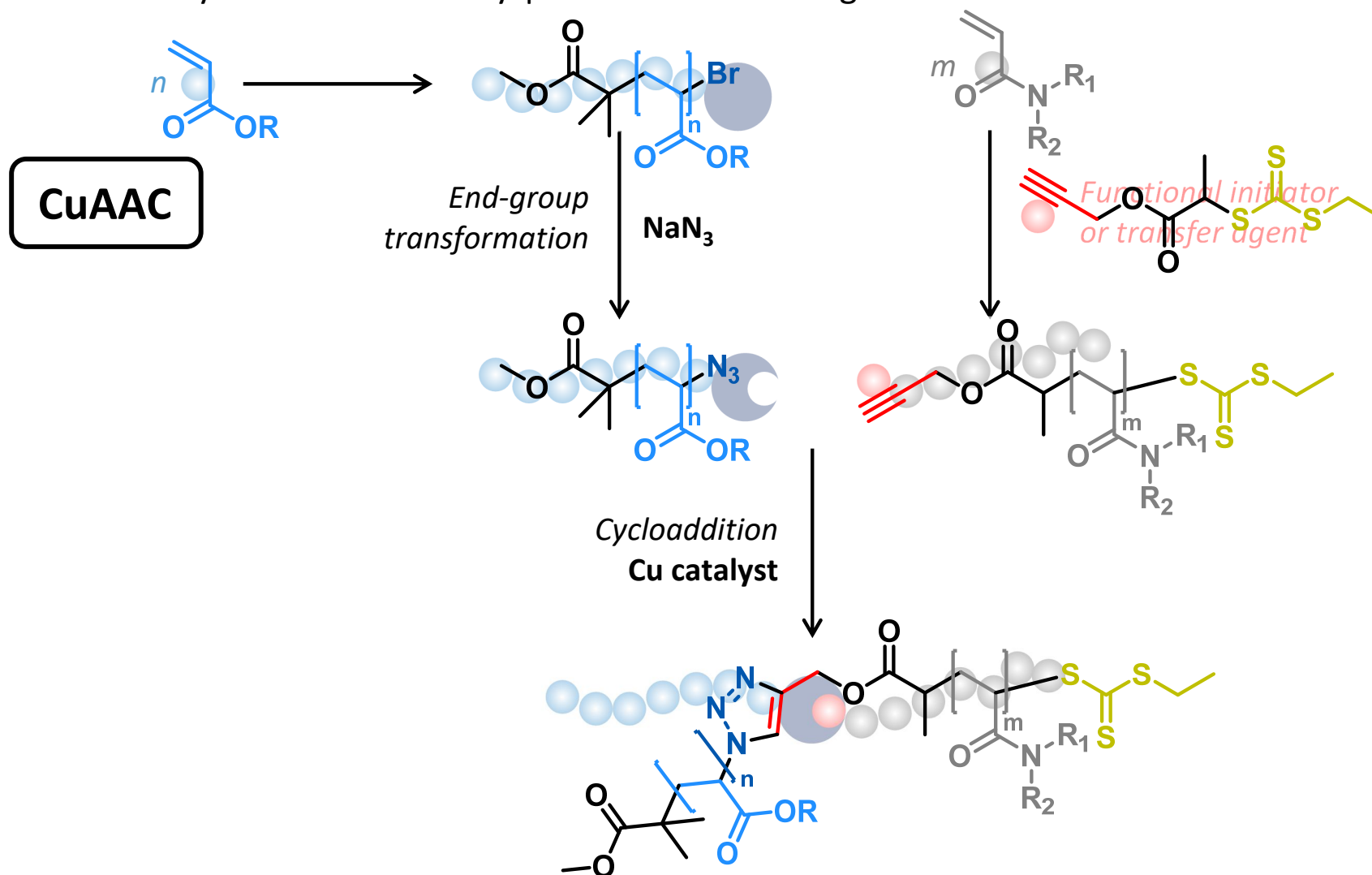


Chem. Commun. **2005**, 57

Synthesis of Linear Block Copolymers

Coupling of Polymer Strands

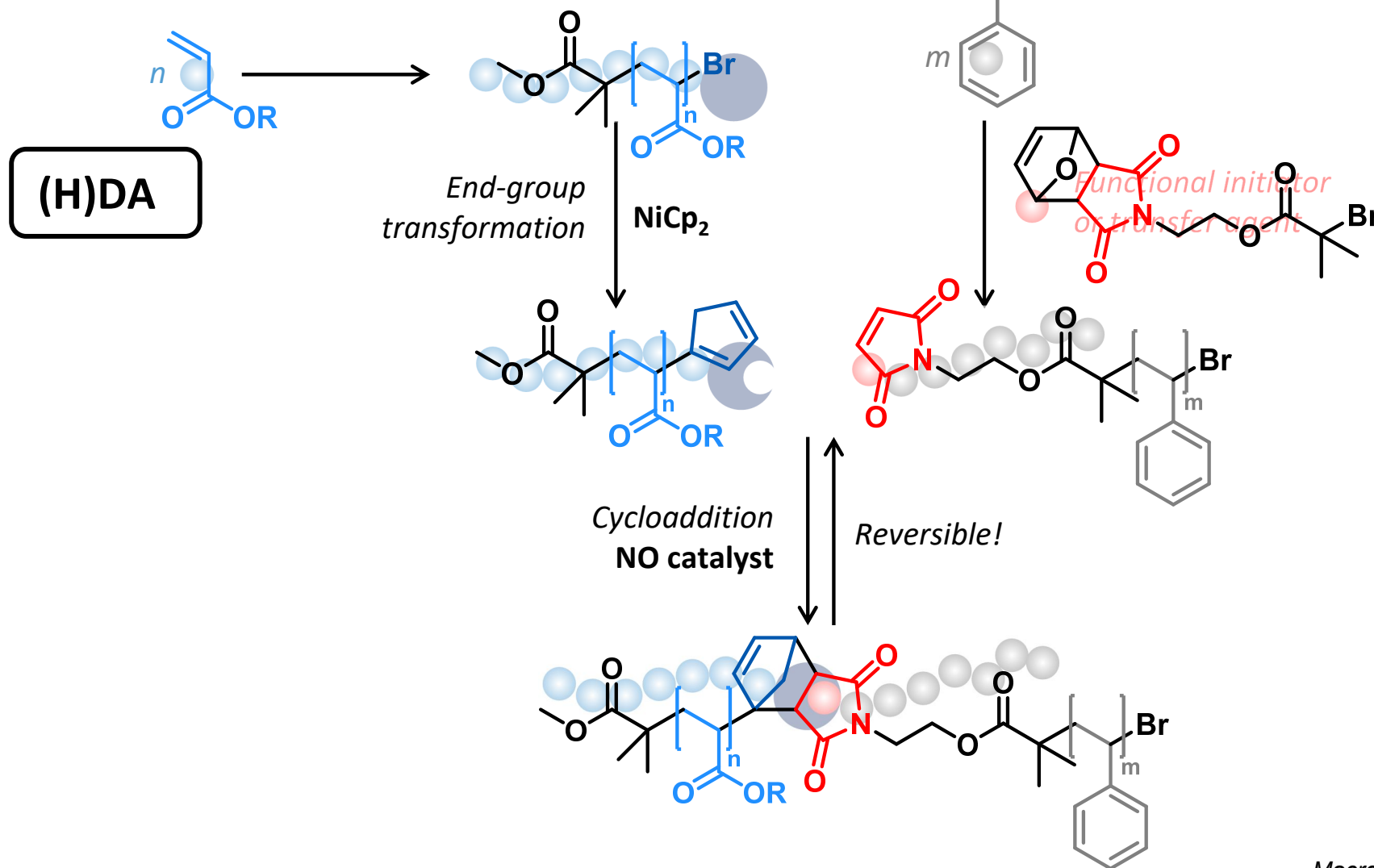
Modern ways: "Click" chemistry | Efficient modular ligations



Synthesis of Linear Block Copolymers

Coupling of Polymer Strands

Modern ways: "Click" chemistry | Efficient modular ligations

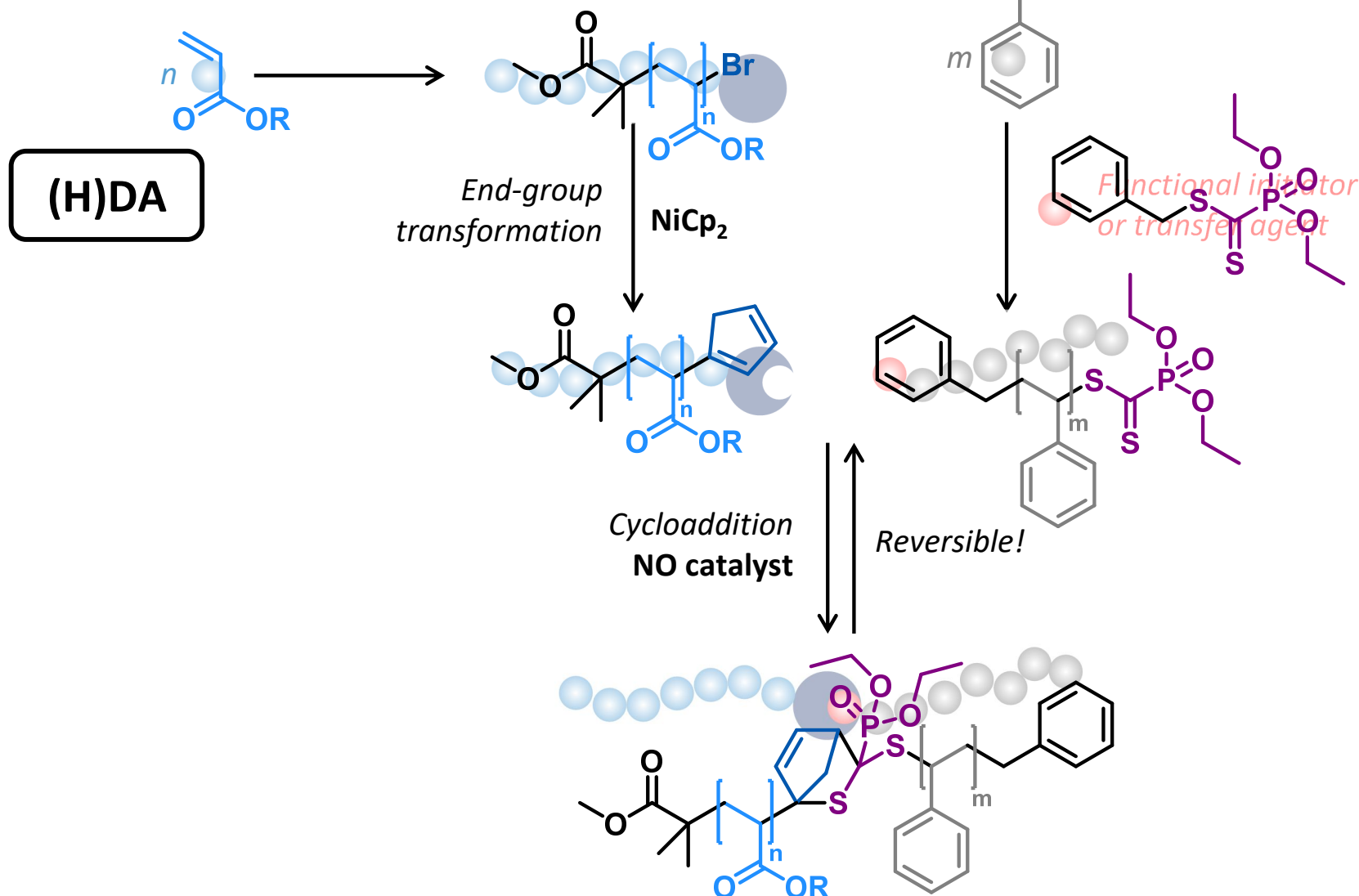


Macromolecules **2010**, 43, 33
Polym. Chem. **2011**, 2, 83

Synthesis of Linear Block Copolymers

Coupling of Polymer Strands

Modern ways: "Click" chemistry | Efficient modular ligations

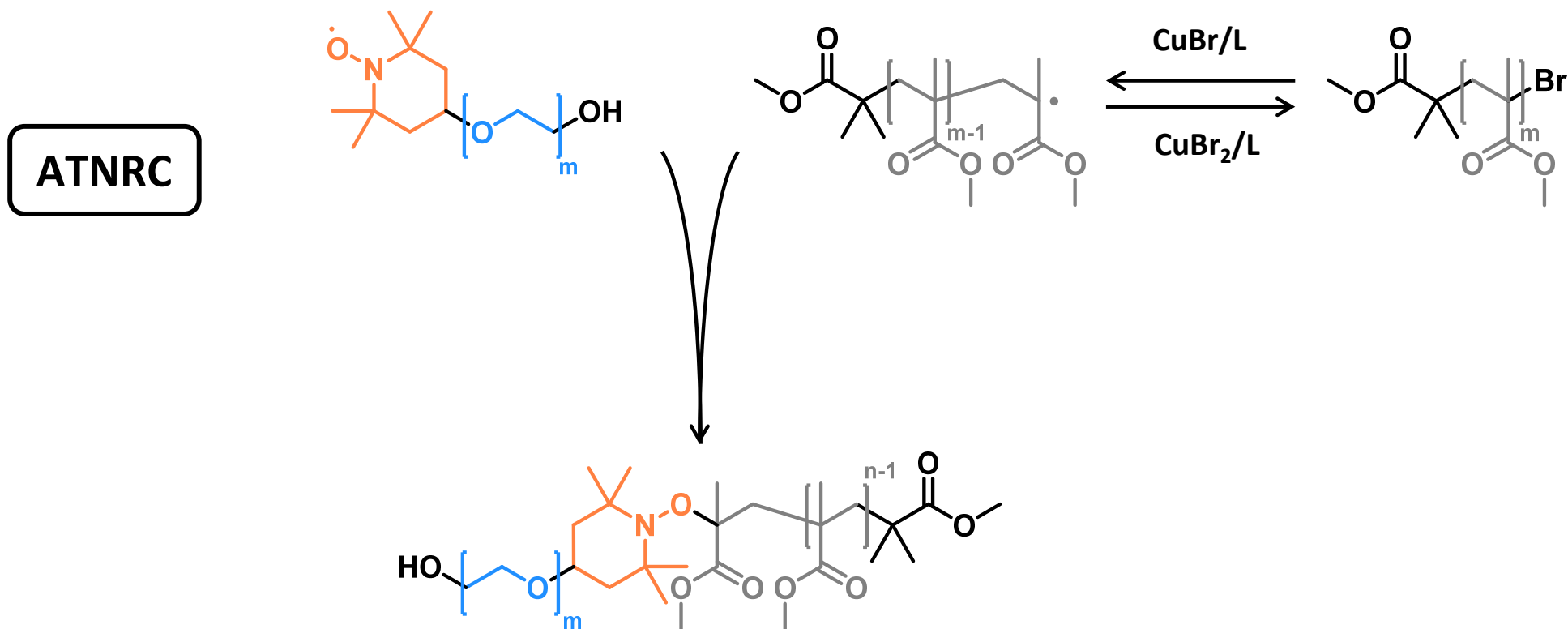


Macromol. Rapid Commun. **2010**, 31, 1247

Synthesis of Linear Block Copolymers

Coupling of Polymer Strands

Modern ways: "Click" chemistry | Efficient modular ligations



Synthesis of Block Copolymers

This is NOT an exhaustive catalogue...

there are many other examples and possible combinations: ∞ ?

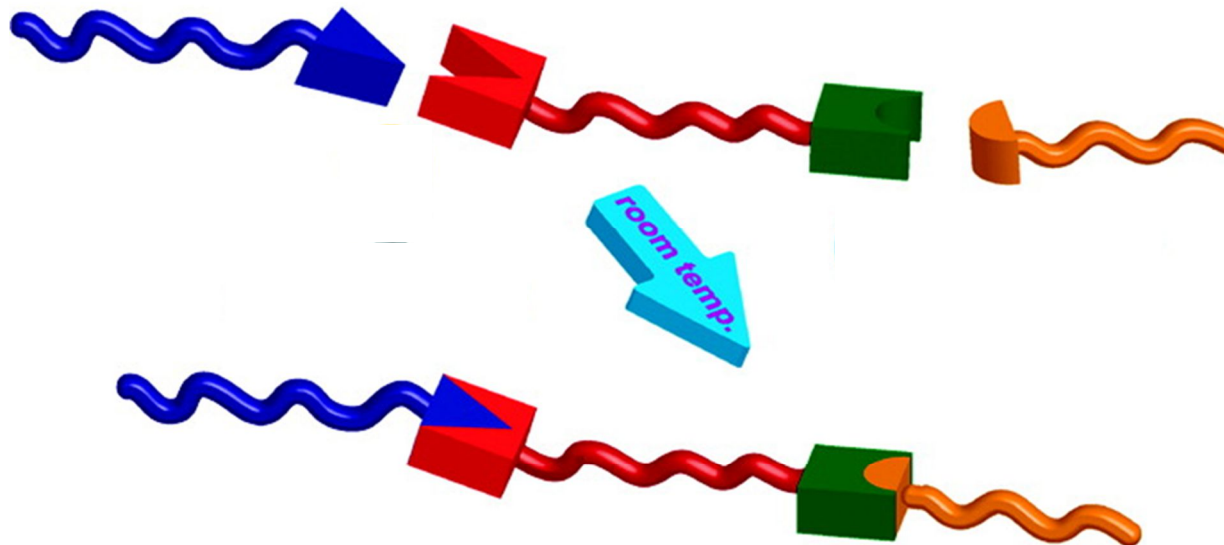
YET, now you have a good overview of the general schemes/routes

Synthesis of Linear Block Copolymers

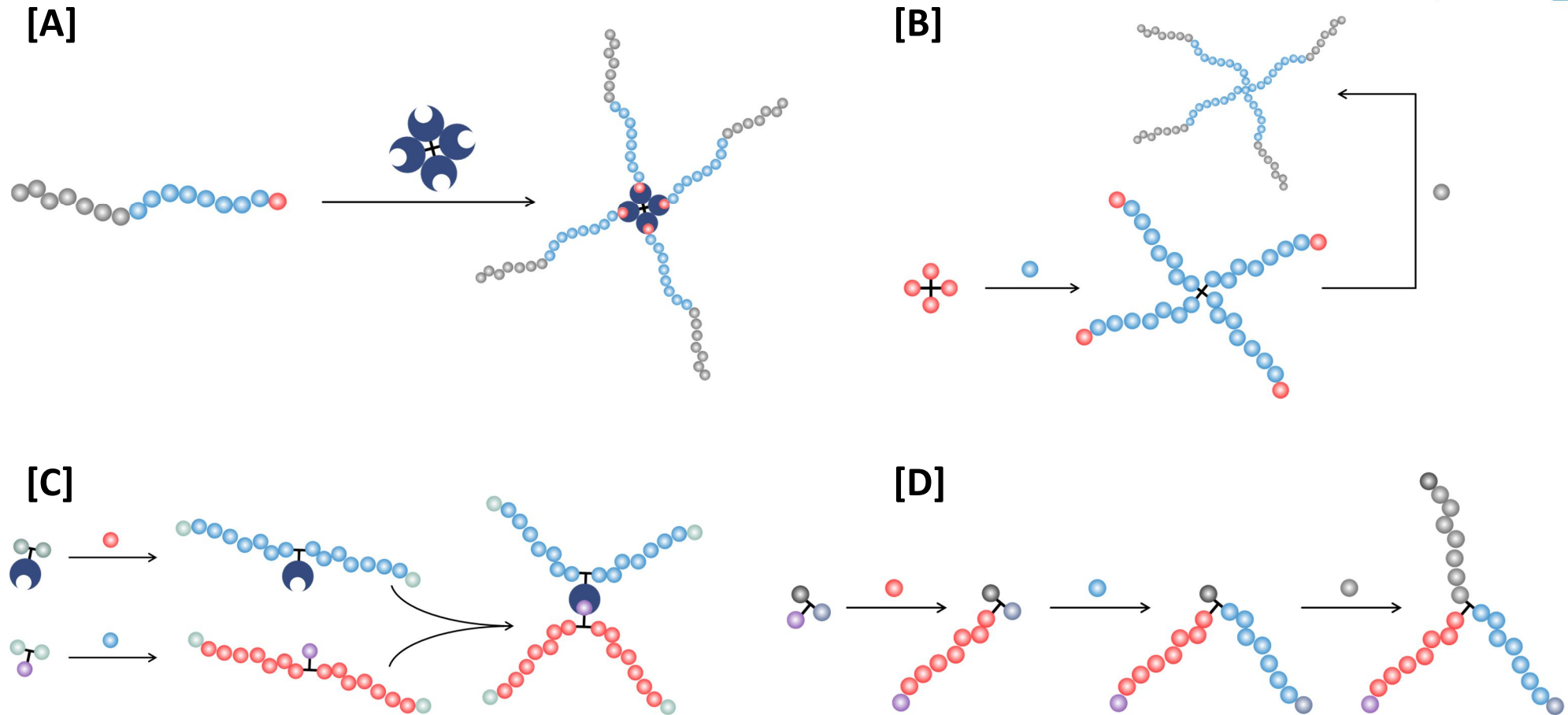
... what about triblock copolymers?

... SAME METHODS!

- * Sequential polymerization
- * Coupling



Synthesis of Star Block Copolymers



Properties of star polymers

(in comparison to linear analogues
of same mass and composition)

→ Lower R_h , R_g , viscosity → Self-assembly properties

Synthesis of Block Copolymers

This is all for the purely synthetic part!

You now see that there are two main strategies:

- * sequential polymerization**
- * coupling of pre-formed polymer chains**
- (* combination of both)**

What are the advantages and drawbacks of each technique?